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NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

Smoking Cessation

Version 1.2026 — April 6, 2026

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NCCN Guidelines for Patients® available at www.nccn.org/patients

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***Peter G. Shields, MD/Chair †**
The Ohio State University Comprehensive Cancer Center - James Cancer Hospital and Solove Research Institute

***Laura J. Bierut, MD/Vice Chair θ**
Siteman Cancer Center at Barnes-Jewish Hospital and Washington University School of Medicine

Sikandar Ansari, MD, MS Ξ
Huntsman Cancer Institute at the University of Utah

David Balis, MD Þ
UT Southwestern Simmons Comprehensive Cancer Center

Lara Coughlin PhD, MS θ
University of Michigan Rogel Cancer Center

James M. Davis, MD Þ
Duke Cancer Institute

Jon Ebbert, MD Þ
Mayo Clinic Comprehensive Cancer Center

Donna Edmondson, MSN, AOCNP, CRNP Ξ
Fox Chase Cancer Center

Sarah Evers-Casey, MPH ¥
Abramson Cancer Center at the University of Pennsylvania

Panagis Galiatsatos, MD, MHS Ξ
Johns Hopkins Kimmel Cancer Center

Brian Hitsman, PhD θ
Robert H. Lurie Comprehensive Cancer Center of Northwestern University

Karen S. Hudmon, DrPH, MS, BS, CTTS Σ
Indiana University Melvin and Bren Simon Comprehensive Cancer Center

Michael T. Jaklitsch, MD ¶
Dana-Farber/Brigham and Women`s Cancer Center | Mass General Cancer Center

Thomas Klingemann, PharmD, CTTS Σ
Fred & Pamela Buffett Cancer Center

Gina Kruse, MD, MPH, MS Þ
University of Colorado Cancer Center

Pamela M. Ling, MD, MPH Þ
UCSF Helen Diller Family Comprehensive Cancer Center

Richard S. Matulewicz, MD, MS, MSCI ω
Memorial Sloan Kettering Cancer Center

Danielle E. McCarthy, PhD θ
University of Wisconsin Carbone Cancer Center

Michael K. Ong, MD, PhD Þ
UCLA Jonsson Comprehensive Cancer Center

Jim Pantelas, BS ¥

Elyse R. Park, PhD, MPH θ
Mass General Cancer Institute

Judith J. Prochaska PhD, MPH θ
Stanford Cancer Institute

Jason D. Robinson, PhD θ
The University of Texas MD Anderson Cancer Center

A. Jossie Sandoval, MD Ξ
City of Hope National Medical Center

Christine E. Sheffer, PhD θ
Roswell Park Comprehensive Cancer Center

Sharon Spencer, MD §
O'Neal Comprehensive Cancer Center at UAB

Marcia M. Tan, PhD, MPH θ
The UChicago Medicine Comprehensive Cancer Center

Tawee Tanvetyanon, MD, MPH †
Moffitt Cancer Center

Hilary A. Tindle, MD, MPH Þ
Vanderbilt-Ingram Cancer Center

Elisa K. Tong, MD, MA Þ
UC Davis Comprehensive Cancer Center

Matthew Triplette, MD, MPH Ξ
Fred Hutchinson Cancer Center

James Urbanic, MD §
UC San Diego Moores Cancer Center

Gregory Videtic, MD §
Case Comprehensive Cancer Center/ University Hospitals Seidman Cancer Center and Cleveland Clinic Taussig Cancer Institute

NCCN

Susan Darlow, PhD
Beth McCullough, RN, BS

Þ Internal medicine
† Medical oncology
¥ Patient advocacy
Σ Pharmacology/ Pharmacogenetics
θ Psychiatry/Psychology/ Behavioral science
Ξ Pulmonary medicine
§ Radiotherapy/ Radiation oncology
¶ Surgery/Surgical oncology
ω Urology
* Discussion Writing Committee Member

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Find an NCCN Member Institution: https://www.nccn.org/home/member-institutions.

NCCN Categories of Evidence and Consensus: All recommendations are category 2A unless otherwise indicated.

See NCCN Categories of Evidence and Consensus.

NCCN Categories of Preference: All recommendations are considered appropriate.

See NCCN Categories of Preference.

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Updates in Version 1.2026 of the NCCN Guidelines for Smoking Cessation include:

Global
- Relapse was revised to: return to smoking.
- Behavior/behavioral therapy was revised to: behavioral treatment.
- Where tobacco is used *or nicotine-containing* was added.

INTRO-1
- Second paragraph revised: *The primary goal for providers should be that their patients achieve complete and sustained abstinence from cigarette smoking.* Non-combustible nicotine-containing products such as e-cigarettes or nicotine pouches are *sometimes* used by people who smoke to quit smoking. ~~*These alternative nicotine products are not FDA approved for smoking cessation. The primary goal for providers should be that their patients achieve complete and sustained abstinence from cigarette smoking.*~~
- Third paragraph revised: As a generalization, the principles for treating tobacco use disorder of any tobacco *or nicotine-containing* products are similar. Therefore, the NCCN Guidelines for Smoking Cessation are directly applicable to not only combustible tobacco products (eg, cigarettes, pipes, cigars, hookah, cigarillos) but also electronic cigarettes (e-cigarettes)/vaping, ~~and~~ smokeless tobacco products*, and oral* ~~(eg,~~ nicotine pouches~~)~~.

INTRO-2
- Bullet moved to first bullet: Treatment for smoking is an integral part of oncology treatment and should be continued throughout the entire oncology care continuum, from treatment, to survivorship, to end-of-life care. . .
- Second bullet, second sub-bullet revised: Behavior*al* ~~therapy~~ *treatment* with multiple counseling sessions is most effective. At a minimum, brief advice is needed. Quitlines are a source of brief behavioral ~~counseling~~ *treatment* for quitting smoking, and *along with pharmacotherapy,* may have added value, especially in lower-resource settings.
- Third bullet revised: *Cancer health care systems should systematically screen all patients for tobacco or nicotine-containing products use.* Smoking status should be documented in the patient health record. . .
- Fourth bullet, last sentence revised: . . .It may take more than one quit attempt with the same therapy to achieve long-term cessation. Health care professionals should educate patients to take smoking ~~treatment~~ *cessation* medications according to the prescribed regimen and for the full course of therapy.

SC-1
- Initial Evaluation:
  - First bullet, new sub-bullet added: (If yes) When was the last time you smoked a cigarette?
  - Bullet removed: Do you currently smoke cigarettes or have you smoked in the last 30 days?
- Status, top option revised: Individuals who ~~currently smoke and/or those who~~ have smoked within the last 30 days → Assessment of individuals who ~~currently~~ *have* smoke*d within last 30 days* (SC-2).
- New footnote f added: Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health (see Practical Concerns). See NCCN Guidelines for Distress Management (DIS-A).

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Updates in Version 1.2026 of the NCCN Guidelines for Smoking Cessation include:

SC-2
- Title revised: Individuals Who *Have* ~~Currently~~ Smoke*d* ~~(~~Within Last 30 Days~~)~~ (Also page SC-3)
- Evaluation:
  - First bullet, first sub-bullet revised: How much *per day* do you currently smoke or use tobacco *or nicotine-containing products* (eg, cigarettes, pipes, cigars, hookah, cigarillos, non-combustible nicotine-containing products, smokeless tobacco, nicotine pouches)~~ per day~~?
  - Second bullet:
    - ◊ First sub-bullet revised: What is the longest period you have gone without smoking *or using nicotine-containing products*?
    - ◊ Second sub-bullet revised: Which strategies were ~~successful~~ *helpful*?
      - – Tertiary bullet revised: Which were ~~unsuccessful~~ *least helpful* and why? . . .
      - – Following Evaluation column, third bullet revised: Assess patient readiness to *set a* quit *date in the next 30 days*

SC-3
- Management:
  - Top column:
    - ◊ Second bullet revised: *Provide counseling and* ~~E~~encourage continued abstinence from smoking.
    - ◊ New bullet added: Continue behavioral treatment and pharmacotherapy as indicated.
  - Following bottom column, option revised: Reassess readiness to quit at each visit *and offer smoking cessation treatment services*.
    - ◊ Arrows added to lead back to Status, Quit within prior 30 days and Ready to set a quit date.
- Footnote p revised: Longer periods of abstinence from smoking confer better surgical outcomes but should not *affect the offer of or* delay appropriate timing of cancer resection. . .
- New footnote s added: Examples are local and national quitline services and Smokefree.gov texting programs.

SC-4
- Significant changes made to this page.
- New footnote t added: Risk factors for returning to smoking include: frequent/intense cravings, elevated anxiety/stress/depression, current or prior history of psychiatric disorders, chronic/uncontrolled pain, living/working with someone who uses tobacco or nicotine-containing products, time since quitting (<1 year), recently initiated quit attempt and/or pharmacotherapy, drug/alcohol use (eg, cannabis, opioids, stimulants), low self-confidence in ability to quit/maintain abstinence, recent history of smoking ≥20 cigarettes per day or time to first cigarette within 30 minutes of waking, and unstable living conditions.

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Updates in Version 1.2026 of the NCCN Guidelines for Smoking Cessation include:

SC-5
- First column:
  - Title revised: Primary Treatment *for Current Quit Attempt*
  - New options added:
    - ◊ or Varenicline + short-acting NRT + behavioral treatment x 12 wks
    - ◊ or Varenicline + nicotine patch + behavioral treatment x 12 wks (category 2B)
- Last column revised: If ~~relapse~~ *return to smoking*, re-initiate primary therapy with either combination NRT *and/*or varenicline.
- Footnotes revised:
  - Footnote bb: For those being treated with varenicline, the addition of NRT or bupropion can be considered as clinically indicated to maintain abstinence despite continued urges to smoke. *Short-acting NRT can be added to varenicline to control/manage cravings. For patients who are heavily dependent and/or have unsuccessfully tried varenicline in the past, varenicline + patch can be considered.* (Also page SC-6)
  - Footnote cc: Most nicotine withdrawal symptoms manifest in the first 1–2 days, peak within the first week, and subside within 2–4 weeks of quitting. Increased appetite, weight gain, ~~and~~ cravings*, and mood changes* may persist for several months. It is important to ~~parse out perceived~~ *distinguish between* withdrawal symptoms from treatment*-related* side effects *in order to not discontinue therapy early*. Encourage continued therapy through brief slips. ~~Hughes JR. Nicotine Tob Res 2007;9:315-327; Benowitz NL. Med Clin North Am 1992;76:415-437.~~ (Also page SC-6)

SC-A (1 of 2)
- General:
  - Third bullet revised: For patients using alternative ~~approaches~~ *nicotine products*, continue to provide support during quit attempts *and encourage the use of evidence-based approaches to avoid delays in achieving abstinence*.
  - Bullet removed: Encourage the use of evidence-based approaches to avoid delays in achieving abstinence.
- Harm Reduction:
  - First bullet, new sentence added: . . .The current list of tobacco products (including e-cigarette brands) approved for legal marketing by the FDA's Center for Tobacco Products is found at: https://www.accessdata.fda.gov/ scripts/searchtobacco/.
  - Second bullet revised: Some of these products have been determined by the FDA's Center for Tobacco Products to be MRTPs. MRTPs have been determined to be less harmful than cigarette smoking *but are not necessarily safe*. . .
  - Fourth bullet revised: *Persistent* ~~D~~*d*ual use of cigarette smoking and alternative tobacco *or nicotine-containing* products may be more harmful than smoking alone and should be actively discouraged.

SC-A (2 of 2)
- E-Cigarettes (Vaping), seventh bullet revised: ~~Optimal methods for vaping cessation are currently unknown. Because of this, p~~*P*roviders are encouraged to follow guidelines for smoking cessation in the treatment of e-cigarette use including behavioral counseling treatment and pharmacotherapy.
- Last bullet, last sentence revised: . . .Vitamin E acetate has been largely eliminated from t*he illicit market and was never reported for detection in commercial* vaping products in the United States.

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Updates in Version 1.2026 of the NCCN Guidelines for Smoking Cessation include:

SC-B (1 of 2)
- First bullet revised: ~~The~~ *Since the* 2014 Surgeon General's Report ~~stated that~~: Sufficient evidence ~~exists~~ *has continued* to support ~~a~~ *the* causal relationship between smoking and adverse health outcomes, *including* increased all-cause ~~mortality~~ and cancer ~~-specific~~ *related* mortality *and morbidity*, and increased risk for secondary primary cancers. *The 2020 Surgeon General's Report on Smoking Cessation and subsequent studies have demonstrated that timely enrollment in evidence-based tobacco treatment programs, particularly within 6 months of a cancer diagnosis, can significantly improve survival outcomes for patients with cancer.*
  - Tertiary bullet removed: Existing evidence is suggestive of a link between smoking and increased risk of cancer recurrence, poor treatment response, and increased treatment-related toxicity.
- Treatment-Specific Risks, third bullet, tertiary bullet revised: ~~Postoperative mortality rates are higher among patients who smoke~~ *Survival outcomes following surgery are worse in patients who smoke*.

SC-B (2 of 2)
- References updated.

SC-C (1 of 3)
- Cancer-Related Resources for Patients:
  - American Society of Clinical Oncology (ASCO), bullet removed: Resource: "Stopping Tobacco Use After a Cancer Diagnosis" https://www.asco.org/sites/new-www.asco.org/files/content-files/ practice-and-guidelines/documents/stopping-tobacco-use-booklet.pdf
- General Resources for Patients:
  - New option added: State/Territory Quitlines and Services
    - ◊ New website added: https://www.naquitline.org/page/mapus

SC-C (3 of 3)
- Cancer-Related Resources for Health Professionals:
  - Third option revised: *American College of Surgeons* Commission on Cancer (~~Just/Beyond ASK~~ *Standard 5.9: Smoking Cessation for Patients with Cancer*): website updated.
  - New option added: Cancer Center Cessation Initiative Tobacco Treatment Roadmap for Cancer and Other Patients
    - ◊ New bullet added: https://ctri.wisc.edu/c3i-roadmap-welcome/

SC-E (1 of 3)
- First bullet revised: Pharmacotherapy is most effective when combined with behavior*al* ~~therapy~~ *treatment*. Brief advice is better than ~~no counseling~~ *nothing*.
- Second bullet, new tertiary bullet added: Contingency management may be considered as an adjunct behavioral treatment when standard counseling and pharmacotherapy alone are insufficient, or when rapid abstinence is clinically important (eg, prior to surgery).
- Fourth bullet revised: Specialized treatment centers ~~may consider providing~~ *should provide* treatment for smoking targeted specifically to patients with cancer (eg, individual therapy and group support that focus on challenges specific to cancer survival and treatment). . .

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SC-E (2 of 3)
- Header revised: Behavior*al* ~~Therapy~~ *and Cognitive Behavioral Treatment*/~~Counseling Recommendations~~
- New header added: Contingency Management
  - New bullets added:
    - ◊ Contingency management is the use of structured financial incentives contingent on biochemically verified abstinence or engagement. It is a behavioral treatment to increase smoking cessation rates.
    - ◊ Contingency management may be particularly helpful for patients with cancer at elevated risk for continued smoking, including those experiencing high stress, depression or anxiety, financial strain, or other social determinants that interfere with treatment engagement.
    - ◊ Contingency management should be delivered using objective biochemical verification, typically expired carbon monoxide (CO) or equivalent cotinine testing. Verification frequency should be highest during the first several weeks after cessation, when risk of return to smoking is greatest.
    - ◊ Incentives should be structured, transparent, and time limited.
    - ◊ Contingency management is most effective when integrated with standard counseling and pharmacotherapy.
    - ◊ Programs should specify the target behavior, verification method, schedule of reinforcement, and duration of contingency management.

SC-E (3 of 3)
- References updated.

SC-F (1 of 4)
- Second bullet:
  - Second tertiary bullet revised: Most nicotine withdrawal symptoms manifest in the first 1–2 days, peak within the first week, and subside within 2–4 weeks of quitting. Increased appetite, weight gain, ~~and~~ cravings*, and mood changes* may persist for several months. It is important to ~~parse out perceived~~ *distinguish between* withdrawal symptoms from treatment-*related* side effects *in order to not discontinue therapy early*. . .
  - Fourth tertiary bullet revised: Pharmacotherapy dose adjustments, including increasing the dosages of NRT and varenicline (but not bupropion), may be considered ~~as clinically indicated~~. *In some cases, dose reduction is warranted based on clinical indicators.*
- Third bullet revised: Track attempts at smoking reduction. If reduction efforts ~~stall~~ *plateau*, or reaching total abstinence seems unlikely, consider switching to a different pharmacotherapy.
- Last bullet revised: When considering pharmacotherapy options, consider cultural and contextual factors including cost, ability to pay, insurance coverage, ability to adhere to the treatment regimen, and patient preferences. *Address patient concerns about pharmacotherapy benefits, myths, and risks.*
- New bullet added: For those being treated with varenicline, the addition of NRT or bupropion can be considered as clinically indicated to maintain abstinence despite continued urges to smoke. Short-acting NRT can be added to varenicline to control/manage cravings. For patients who are heavily dependent and/or have unsuccessfully tried varenicline in the past, varenicline + patch can be considered.
- Pharmacotherapy Options:
  - Other Recommended:
    - ◊ First bullet revised: Varenicline/*short-acting* NRT
    - ◊ Second bullet revised: Bupropion *(eg, for patients with depression or fatigue)*
    - ◊ New bullet added: Varenicline/nicotine patch (category 2B)

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Updates in Version 1.2026 of the NCCN Guidelines for Smoking Cessation include:

SC-F (2 of 4):
- Standard Dose/Administration:
  - Combination NRT (preferred)
    - Second bullet revised: Begin with ~~21-mg~~ *nicotine* patch + short-acting NRT.
    - New sub-bullets added:
      - If smoking 1-10 cigarettes/day, use 14-mg patch.
      - If smoking >10 cigarettes/day, use 21-mg patch.
        - Tertiary bullet revised: If ~~21-mg~~ *initial* patch is not effective, consider ~~using more than one patch to increase~~ *increasing* the dose, ~~to 35 or 42 mg~~ *including up to more than one patch.*
  - Bupropion, third bullet revised: Day 4 ~~-~~ *through* end of treatment:
- Duration revised: Minimum of 3 months (6 months *or longer* preferred) of pharmacotherapy is recommended.

SC-F (3 of 4)
- Adverse Effects and Contraindications:
  - Varenicline (preferred), second bullet revised: Although t~~hese side effects are uncommon~~ *the literature does not support that it increases the risk for neuropsychiatric issues*, health care professionals should monitor for the development or worsening of serious neuropsychiatric issues (ie, depression and suicidal ideation/behavior), including for patients without a previous history, and discontinue use if these signs occur.
  - Bupropion, fourth bullet revised: ~~Bupropion inhibits CYP2D6 and can increase concentrations of antidepressants, antipsychotics, beta-blockers, tamoxifen, and type 1C antiarrhythmics.~~ *Inhibitors of CYP2B6 (eg, ticlopidine, clopidogrel) can increase bupropion (active drug) and require dose adjustments.* Consider dose reduction when using with bupropion. Use bupropion with caution when used with drugs that lower seizure threshold. Caution should be used for bupropion for patients treated with tamoxifen. Central nervous system (CNS) toxicity can occur when bupropion is used concomitantly with dopaminergic drugs *(eg, amantadine, levodopa)*.

SC-F (4 of 4)
- New reference added: Herd N, Borland R. The natural history of quitting smoking: findings from the International Tobacco Control (ITC) Four Country Survey. Addiction 2009;104:2075-2087.

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GENERAL PRINCIPLES OF THE SMOKING CESSATION GUIDELINES

The NCCN Panel recommends that treatment plans are offered for all people with cancer who smoke, regardless of readiness to quit. Treatment plans should include:

- evidence-based motivational strategies and behavioral treatment,
- evidence-based pharmacotherapy
- close follow-up with retreatment as needed, and
- recognition that there is a continuum of harm for nicotine-containing products.

The primary goal for providers should be that their patients achieve complete and sustained abstinence from cigarette smoking. Non-combustible nicotine-containing products such as e-cigarettes or nicotine pouches are sometimes used by people who smoke to quit smoking. These alternative nicotine products are not FDA approved for smoking cessation.

As a generalization, the principles for treating tobacco use disorder of any tobacco or nicotine-containing products are similar. Therefore, the NCCN Guidelines for Smoking Cessation are directly applicable to not only combustible tobacco products (eg, cigarettes, pipes, cigars, hookah, cigarillos) but also electronic cigarettes (e-cigarettes)/vaping, smokeless tobacco products, and oral nicotine pouches.

These guidelines are focused on recommendations for the treatment of smoking for patients with cancer and cancer survivors. There are health benefits to smoking cessation even after a cancer diagnosis, regardless of site, stage, or prognosis, namely improvement in cancer treatment outcomes, primary cancer recurrence, and secondary cancers. It is never too late for patients with cancer to quit smoking and experience health benefits. Tobacco use disorder is a chronic relapsing condition.

Note: All recommendations are category 2A unless otherwise indicated.

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GENERAL PRINCIPLES OF THE SMOKING CESSATION GUIDELINES

Clinical Recommendations:

- Treatment for smoking is an integral part of oncology treatment and should be continued throughout the entire oncology care continuum, from treatment, to survivorship, to end-of-life care. An emphasis should be put on patient preferences and values when considering the best approach to smoking treatment during end-of-life care.
- Combining pharmacologic therapy and behavioral treatment is the most effective approach for achieving and maintaining abstinence from smoking, or smoking reduction for those not ready to quit.
  - The most effective pharmacotherapy approaches are combination nicotine replacement therapy (NRT)[a] and/or varenicline.
  - Behavioral treatment with multiple counseling sessions is most effective. At a minimum, brief advice is needed. Quitlines are a source of brief behavioral treatment for quitting smoking, and along with pharmacotherapy, may have added value, especially in lower-resource settings.
- Cancer health care systems should systematically screen all patients for tobacco or nicotine-containing products use. Smoking status should be documented in the patient health record. Patient health records should be updated at regular intervals to indicate changes in smoking status, quit attempts made, and interventions utilized. All patients undergoing smoking treatment should be considered for lung cancer screening.[b]
- Return to smoking and brief slips are common and can be managed. Health care professionals, including tobacco treatment specialists, should discuss this and provide guidance and support to encourage continued abstinence from smoking. Smoking slips are not necessarily an indication to try an alternative method. It may take more than one quit attempt with the same therapy to achieve long-term cessation. Health care professionals should educate patients to take smoking cessation medications according to the prescribed regimen and for the full course of therapy.
- There is a continuum of harm from most harmful (cigarettes and other combustible tobacco products) to less harmful non-combustible nicotine-containing products (eg, e-cigarettes, heat-not-burn products, snus, and nicotine pouches). There is growing evidence from randomized clinical trials and meta-analyses that e-cigarettes are effective tools for smoking cessation. However, e-cigarettes are addictive, and they contain and produce harmful chemicals. Providers should strongly support the use of evidence-based pharmacotherapies for smoking cessation. However, for patients who choose to use alternative nicotine products (non-combustible) for smoking cessation, the principal focus is to support complete abstinence from cigarette smoking. Some alternative tobacco products are designated by the FDA's Center for Tobacco Products as modified risk tobacco products (MRTPs), but this is not the same as the FDA approval for the treatment of smoking.

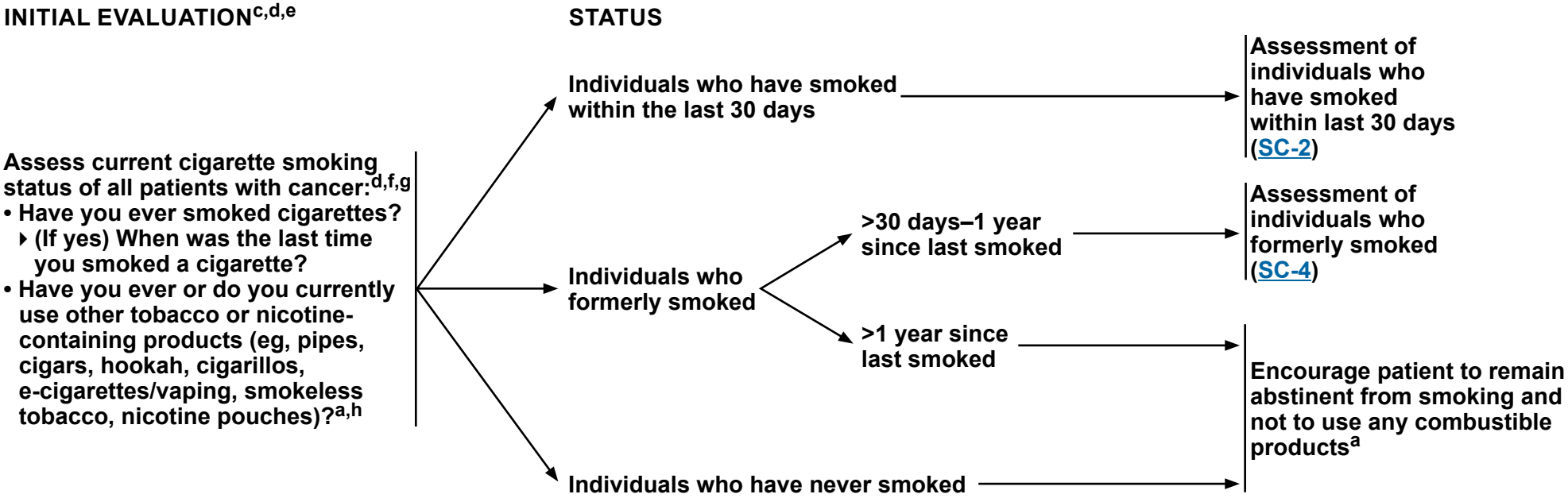
[a] Combination NRT = Long-acting NRT (nicotine patch) + short-acting NRT (ie, lozenge, gum, nasal spray).
[b] For lung cancer screening recommendations, see NCCN Guidelines for Lung Cancer Screening.

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EVALUATION AND ASSESSMENT OF PATIENT SMOKING[a,b]



[a] The Smoking Cessation algorithms can be applied for any other tobacco or nicotine-containing product. Patients with cancer should be encouraged to achieve and maintain abstinence from all combustible tobacco products (eg, cigarettes, pipes, cigars, hookah, cigarillos) and alternative nicotine products. For information about e-cigarettes and other non-combustible nicotine-containing products, see Principles of Alternative Approaches to Treatment of Smoking (SC-A).
[b] General Principles of the Smoking Cessation Guidelines.
[c] For lung cancer screening recommendations, see NCCN Guidelines for Lung Cancer Screening.
[d] Initial evaluation and assessment of patient smoking may be completed by any member of the health care team, including physicians, nurses, medical assistants, health educators, pharmacists, tobacco treatment specialists, or other dedicated staff.
[e] Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health (see Practical Concerns). See NCCN Guidelines for Distress Management (DIS-A).
[f] Smoking status of all tobacco products, including e-cigarettes and other non-combustible nicotine-containing products, should be documented in the patient health record and assessment should be repeated at every visit.
[g] Treatment for smoking should be offered to all people with cancer who smoke regardless of cancer stage or prognosis. See Smoking-Associated Risks for Patients With Cancer (SC-B).
[h] Patients who use multiple tobacco- or nicotine-containing products may have more difficulty achieving abstinence.

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INDIVIDUALS WHO SMOKED WITHIN LAST 30 DAYS

EVALUATION

Assess and document in the patient health record:

- **Tobacco use:**
  - **How much per day do you currently smoke or use tobacco or nicotine-containing products (eg, cigarettes, pipes, cigars, hookah, cigarillos, non-combustible nicotine-containing products, smokeless tobacco, nicotine pouches)?**
  - **What is the typical amount?**
  - **How soon do you smoke or use tobacco or nicotine-containing products after you wake up? (ie, within 30 minutes)[i]**
- **History of quit attempts:**
  - **What is the longest period you have gone without smoking or using nicotine-containing products?**
  - **Which strategies were helpful?**
    - ◊ **Which were least helpful and why? (eg, side effects, cost, continued cravings, did not work, inadequate behavioral support)**
  - **When was your last quit attempt?**
  - **Did you use anything to help you quit in the past? If so, what?**
    - ◊ **Medications[j] (eg, varenicline, bupropion, NRT)**
    - ◊ **Support group**
    - ◊ **Behavioral treatment**
    - ◊ **Quitlines, websites, smart phone apps, or other media**
    - ◊ **Non-combustible nicotine-containing products[k]**
    - ◊ **Nicotine pouches**
    - ◊ **Other**
- **Tobacco or nicotine-containing product use of individuals within household and other shared spaces[l]**

→ **Advise patients to quit[m]**
- **Review risks of smoking and benefits of quitting (Smoking-Associated Risks for Patients With Cancer [SC-B])**
- **Provide patient education resources (Smoking Cessation and Treatment Resources for Patients/Health Care Professionals, [SC-C])**
- **Assess patient readiness to set a quit date in the next 30 days[n]**

→ **Management (SC-3)**

[i] Time to first cigarette is used to select the strength of the nicotine gum and lozenge (2 mg, 4 mg).

[j] Document medication name, strength, dose, and duration of use during previous quit attempts. If medication was discontinued early during prior quit attempt(s), longer duration of use may improve efficacy.

[k] Principles of Alternative Approaches to Treatment of Smoking (SC-A).

[l] Smoking by other household members is an important predictor for not remaining abstinent and exposes the patient to secondhand smoke. To benefit the patient, individuals in the household and other shared spaces should be encouraged to abstain from smoking; resources may be provided.

[m] Brief advice of about 3 minutes by health care professionals is associated with a small but important increase in smoking abstinence rates.

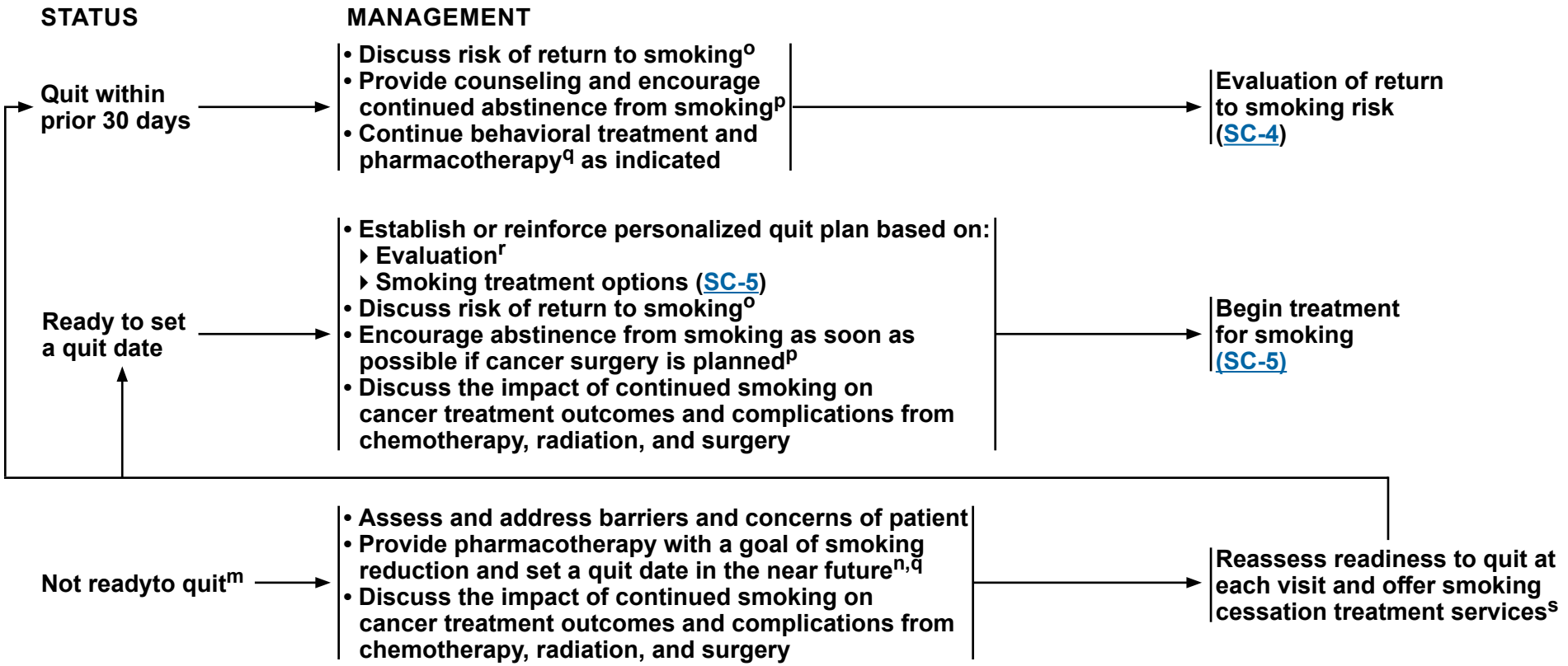
[n] Making an immediate quit attempt is recommended, but smoking reduction with pharmacotherapy and counseling may be considered with a goal of achieving and maintaining abstinence from smoking at a future quit date (ie, 1–3 mo).

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INDIVIDUALS WHO HAVE SMOKED WITHIN LAST 30 DAYS



[m] Brief advice of about 3 minutes by health care professionals is associated with a small but important increase in smoking abstinence rates.

[n] Making an immediate quit attempt is recommended, but smoking reduction with pharmacotherapy and counseling may be considered with a goal of achieving and maintaining abstinence at a future quit date (ie, 1–3 mo).

[o] Health care professionals, including tobacco treatment specialists, should discuss risks of return to smoking and slips and provide guidance and support to encourage continued abstinence from smoking. See Principles of Behavioral Treatment of Smoking (SC-E).

[p] Longer periods of abstinence from smoking confer better surgical outcomes but should not affect the offer of or delay appropriate timing of cancer resection. See Principles of Smoking Cessation and Cancer Surgery (SC-D).

[q] Principles of Pharmacotherapy for Smoking (SC-F).

[r] Adjustments to therapy length, intensity, and surveillance may be considered, as clinically indicated, for patients with high nicotine use and/or prior unsuccessful quit attempts.

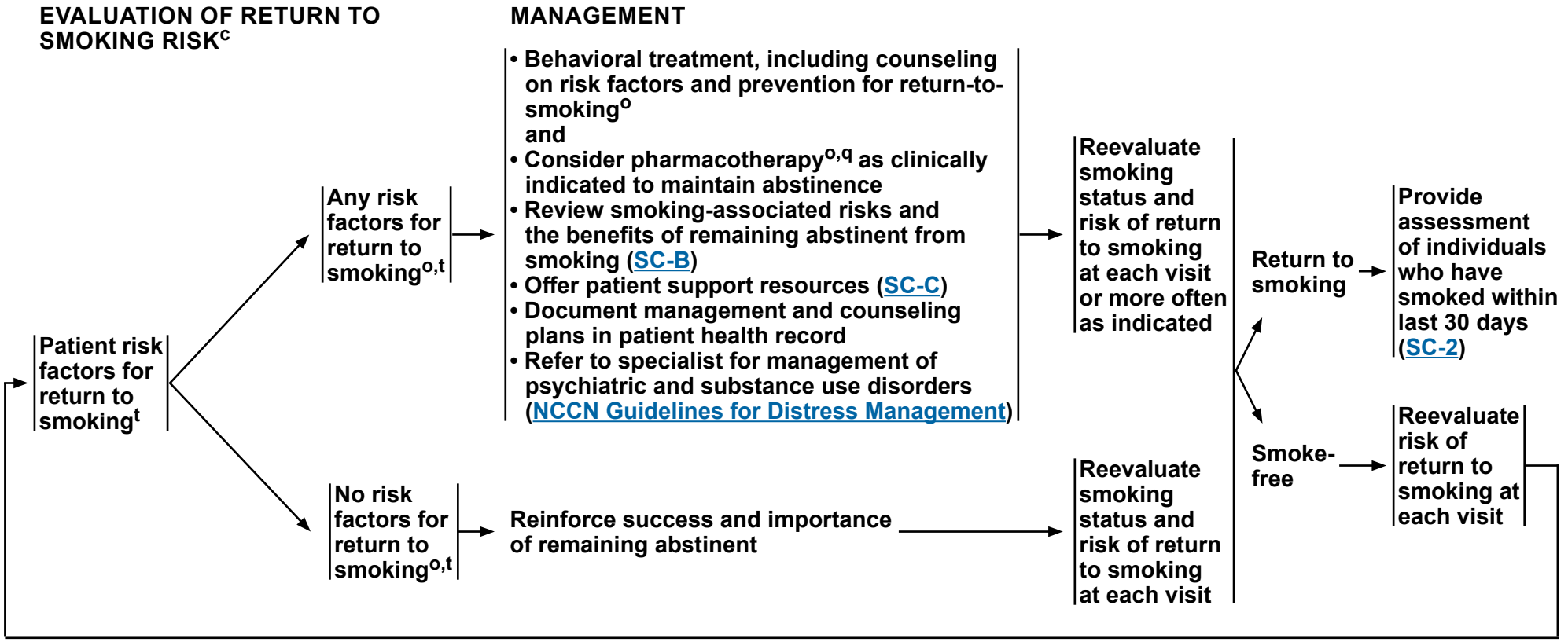
[s] Examples are local and national quitline services and Smokefree.gov texting programs.

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INDIVIDUALS WHO FORMERLY SMOKED (More Than 30 Days Since Last Smoked)



[c] For lung cancer screening recommendations, see NCCN Guidelines for Lung Cancer Screening.
[o] Health care professionals, including tobacco treatment specialists, should discuss risks of return to smoking and slips and provide guidance and support to encourage continued abstinence from smoking. See Principles of Behavioral Treatment of Smoking (SC-E).
[q] Principles of Pharmacotherapy for Smoking (SC-F).
[t] Risk factors for returning to smoking include: frequent/intense cravings, elevated anxiety/stress/depression, current or prior history of psychiatric disorders, chronic/uncontrolled pain, living/working with someone who uses tobacco or nicotine-containing products, time since quitting (<1 year), recently initiated quit attempt and/or pharmacotherapy, drug/alcohol use (eg, cannabis, opioids, stimulants), low self-confidence in ability to quit/maintain abstinence, recent history of smoking ≥20 cigarettes per day or time to first cigarette within 30 minutes of waking, and unstable living conditions.

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GENERAL APPROACH TO TREATMENT OF SMOKING FOR PATIENTS WITH CANCER AND SURVIVORS

PRIMARY TREATMENT FOR CURRENT QUIT ATTEMPT[k,q,v,w,x]

Combination NRT[u,q,y] + behavioral treatment x 12 wks[z] (category 1)
or
Varenicline[q,aa,bb] + behavioral treatment x 12 wks[z] (category 1)
or
Varenicline[q,aa,bb] + short-acting NRT[q] + behavioral treatment x 12 wks[z]
or
Varenicline[q,aa,bb] + nicotine patch[q] + behavioral treatment x 12 wks[z] (category 2B)

→ **ASSESSMENT/FOLLOW-UP**

Assess effectiveness of counseling and pharmacotherapy, smoking status, and adverse effects of pharmacotherapy in-person or by phone/telehealth at a minimum:[cc,dd,ee,ff]
- **Within 3 wks of initiating treatment for smoking (within 1 wk preferred) and**
- **Following 12 wks of therapy**

→ **Smoke free** →
- **Motivational strategies for continued abstinence**
- **Extend duration of pharmacotherapy as clinically indicated to maintain abstinence (ie, 6 mo or longer)**
- **Reassess smoking status at the end of therapy (if past 12 wks) and at 6 & 12 mo after therapy**

→ **If return to smoking, re-initiate primary therapy with either combination NRT and/or varenicline**

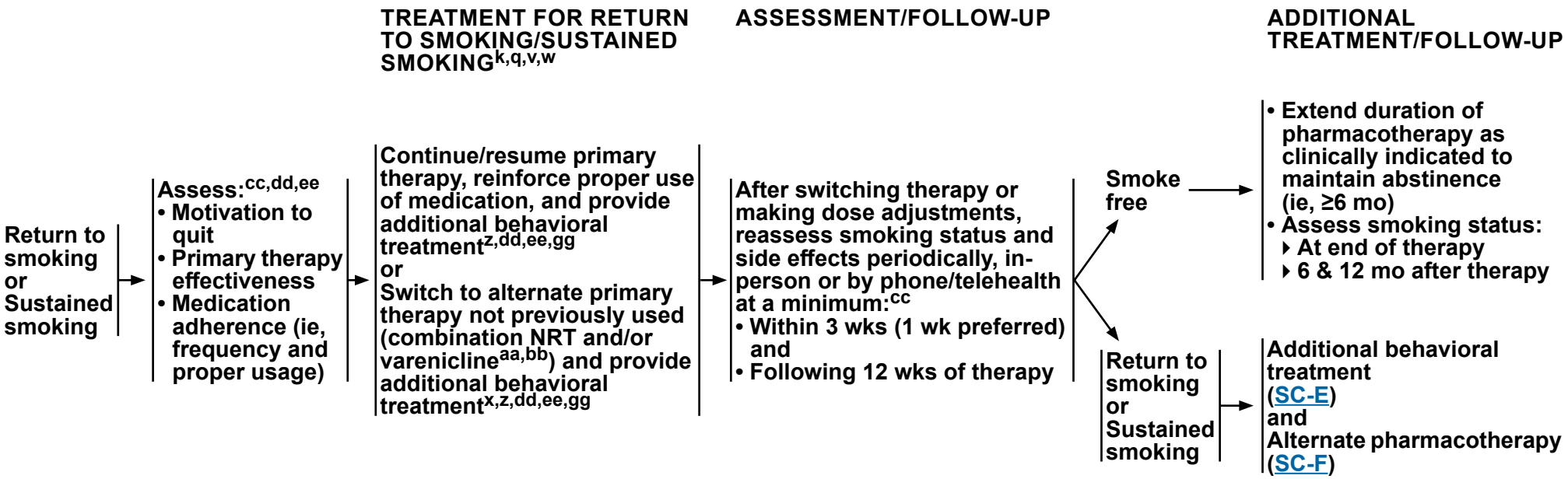
→ **Return to smoking or sustained smoking[n]** → **Return to smoking or sustained smoking (SC-6)**

[k] Principles of Alternative Approaches to Treatment of Smoking (SC-A).
[n] Making an immediate quit attempt is recommended, but smoking reduction with pharmacotherapy and counseling may be considered with a goal of achieving and maintaining abstinence at a future quit date (ie, 1–3 mo).
[q] Principles of Pharmacotherapy for Smoking (SC-F).
[u] Combination NRT = Long-acting NRT (nicotine patch) + short-acting NRT (ie, lozenge, gum, nasal spray).
[v] Principles of Behavioral Treatment of Smoking (SC-E).
[w] The use of cannabis, or other substances associated with return to smoking, including alcohol, is discouraged for those attempting to quit smoking. Lynch KL, et al. Nicotine Tob Res 2019;21:1058-1064.
[x] Combination NRT or varenicline are the preferred pharmacotherapy options. However, bupropion ± NRT may be appropriate for select patients (eg, for those with depression or fatigue). See SC-F.
[y] Patient assistance programs may be available.
[z] Four or more sessions of individual/group therapy is recommended, with additional sustained counseling as needed. Brief advice by health care professionals is associated with a small but important increase in smoking abstinence rates. Longer, more frequent sessions with trained counselors are associated with higher abstinence rates. See Principles of Behavioral Treatment of Smoking (SC-E).
[aa] Nausea is a common side effect of varenicline and may need to be managed for patients with cancer, especially during chemotherapy.
[bb] For those being treated with varenicline, the addition of NRT or bupropion can be considered as clinically indicated to maintain abstinence despite continued urges to smoke. Short-acting NRT can be added to varenicline to control/manage cravings. For patients who are heavily dependent and/or have unsuccessfully tried varenicline, varenicline + patch can be considered.
[cc] Most nicotine withdrawal symptoms manifest in the first 1–2 days, peak within the first week, and subside within 2–4 weeks of quitting. Increased appetite, weight gain, cravings, and mood changes may persist for several months. It is important to distinguish between withdrawal symptoms from treatment-related side effects in order to not discontinue therapy early. Encourage continued therapy through brief slips.
[dd] Adjust behavioral treatment frequency as needed.
[ee] Pharmacotherapy dose adjustments can be considered as clinically indicated. Doses may be reduced for adverse effects or increased if risk of return to smoking is high. See Principles of Pharmacotherapy for Smoking (SC-F).
[ff] Drug levels of certain medications may be altered when a person stops smoking; assess for drug interactions.

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GENERAL APPROACH TO TREATMENT OF SMOKING FOR PATIENTS WITH CANCER AND SURVIVORS
Return to Smoking or Sustained Smoking



k Principles of Alternative Approaches to Treatment of Smoking (SC-A).
q Principles of Smoking Cessation Pharmacotherapy (SC-F).
v Principles of Behavioral Treatment of Smoking (SC-E).
w The use of cannabis, or other substances associated with return to smoking, including alcohol, is discouraged for those attempting to quit smoking. Lynch KL, et al. Nicotine Tob Res 2019;21:1058-1064.
x Combination NRT or varenicline are the preferred pharmacotherapy options. However, bupropion ± NRT may be appropriate for select patients (eg, for those with depression or fatigue). See SC-F.
z Four or more sessions of individual/group therapy is recommended, with additional sustained counseling as needed. Brief advice by health care professionals results in a small but important increase in quit rates. Longer, more frequent sessions with trained counselors are linked to higher success rates. See Principles of Behavioral Treatment of Smoking (SC-E).
aa Nausea is a common side effect of varenicline and may need to be managed for patients with cancer, especially during chemotherapy.
bb For those being treated with varenicline, the addition of NRT or bupropion can be considered as clinically indicated to maintain abstinence despite continued urges to smoke. Short-acting NRT can be added to varenicline to control/manage cravings. For patients who are heavily dependent and/or have unsuccessfully tried varenicline, varenicline + patch can be considered.
cc Most nicotine withdrawal symptoms manifest in the first 1–2 days, peak within the first week, and subside within 2–4 weeks of quitting. Increased appetite, weight gain, cravings, and mood changes may persist for several months. It is important to distinguish between withdrawal symptoms from treatment-related side effects in order to not discontinue therapy early. Encourage continued therapy through brief slips.
dd Adjust behavioral treatment frequency as needed.
ee Pharmacotherapy dose adjustments can be considered as clinically indicated. Doses may be reduced for adverse effects or increased if risk of return to smoking is high. See Principles of Pharmacotherapy for Smoking (SC-F).
gg Decision to continue or switch therapy should be based on prior cessation success, patient preference, toxicity, and/or a change in clinical status (eg, upcoming surgery).

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PRINCIPLES OF ALTERNATIVE APPROACHES TO TREATMENT OF SMOKING[a]

General

- Offer motivational and behavioral support to all patients attempting to quit smoking, regardless of what smoking treatment method(s) is/are being used. See Principles of Behavioral Treatment of Smoking (SC-E).
- For patients who are unable or unwilling to quit, one potential strategy is smoking reduction with a goal of quitting (SC-2).
- For patients using alternative nicotine products, continue to provide support during quit attempts and encourage the use of evidence-based approaches to avoid delays in achieving abstinence.
- Return to smoking and smoking slips are common. Remind patients that repeated attempts with evidence-based methods are frequently needed to achieve longer-term abstinence.

Harm Reduction

- There are a wide variety of products on the market that may be used as alternative nicotine products such as non-combustible tobacco products (smokeless tobacco in the form of snus) and nicotine-containing products without tobacco (nicotine pouches). The current list of tobacco products (including e-cigarette brands) approved for legal marketing by the FDA's Center for Tobacco Products is found at: https://www.accessdata.fda.gov/scripts/searchtobacco/.
- Some of these products have been determined by the FDA's Center for Tobacco Products to be MRTPs. MRTPs have been determined to be less harmful than cigarette smoking but are not necessarily safe. The current list of MRTPs is found at: https://www.fda.gov/tobacco-products/advertising-and-promotion/modified-risk-granted-orders. MRTPs are not approved for smoking cessation by the FDA's Center for Drug Evaluation and Research (CDER).
- The long-term health effects of MRTPs remain unknown, but complete transition from cigarette smoking to MRTPs may reduce smoking-related cancer and disease.
- Persistent dual use of cigarette smoking and alternative tobacco- or nicotine-containing products may be more harmful than smoking alone and should be actively discouraged.
- For patients who choose to use non-combustible alternative products for smoking cessation, the provider's main focus should be to support complete long-term smoking abstinence.

[a] See Discussion for references.

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PRINCIPLES OF ALTERNATIVE APPROACHES TO TREATMENT OF SMOKING[a]

E-Cigarettes (Vaping)

- E-cigarettes are non-combustible tobacco products that deliver nicotine through heated aerosols for inhalation. The use of e-cigarettes is often referred to as vaping.
- The FDA may authorize the sale of some e-cigarettes, but no e-cigarette has been approved as a smoking cessation device.
- There is accumulating evidence from randomized clinical trials and systematic reviews that e-cigarettes are effective for smoking cessation.
- E-cigarettes are addictive, and they contain and produce harmful chemicals.
- If a patient has already quit cigarette smoking through the use of vaping, providers should support continued smoking abstinence. As the patient becomes more confident about maintaining long-term smoking abstinence, the provider can encourage cessation of e-cigarettes, but not at the risk of return to cigarette smoking.
- The long-term health effects of vaping are unknown. However, e-cigarette aerosols contain substantially lower concentrations of harmful toxicants than combustible cigarette smoke.
- Providers are encouraged to follow guidelines for smoking cessation in the treatment of e-cigarette use including behavioral treatment and pharmacotherapy.
- Resources are available to aid in vaping cessation (SC-C 2 of 3).
- EVALI (e-cigarette or vaping product use-associated lung injury) has occurred in patients who have vaped products contaminated with vitamin E acetate added with illicit drug use to e-liquids. These were not commercial products, for example, as approved for marketing by the FDA. Vitamin E acetate has been largely eliminated from the illicit market and was never reported for detection in commercial vaping products in the United States.

Other Alternative Methods

- There is currently insufficient evidence to support the use of alternative methods (eg, hypnosis, acupuncture, nutritional supplements) when used alone or in combination with standard smoking treatment approaches.
- Prior unsuccessful quit attempts with conventional therapies do not justify the use of unproven alternative approaches, because multiple attempts with evidence-based methods may be necessary to achieve abstinence.
- There are very limited, low-quality data regarding the efficacy of exercise-based interventions.

[a] See Discussion for references.

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SMOKING-ASSOCIATED RISKS FOR PATIENTS WITH CANCER

- Since the 2014 Surgeon General's Report[1]: Sufficient evidence has continued to support the causal relationship between smoking and adverse health outcomes, including increased all-cause and cancer-related mortality and morbidity, and increased risk for secondary primary cancers. The 2020 Surgeon General's Report on Smoking Cessation and subsequent studies[2] have demonstrated that timely enrollment in evidence-based tobacco treatment programs, particularly within 6 months of a cancer diagnosis, can significantly improve survival outcomes for patients with cancer.
- Health care professionals should:
  - Inform patients of the potential benefits of abstinence from smoking, including improved survival, treatment outcomes, and health-related quality of life, as well as decreased treatment-related toxicity, drug side effects, and surgical complications.
  - Educate patients on the specific risks of smoking during treatment for their particular cancer.
  - Encourage abstinence from smoking as far in advance as possible before initiating cancer treatment.
  - Consider patient smoking status prior to initiating treatment and when making decisions regarding treatment selection, dosage, and timing of initiation.
  - Discuss treatment for smoking with all patients who smoke. Patient satisfaction is enhanced when smoking treatment is offered by health care professionals.
  - To benefit the patient, individuals in the household and other shared spaces should be encouraged to abstain from smoking; resources may be provided.[3]

Treatment-Specific Risks (see Discussion for additional information)

- Smoking can impact the metabolism of chemotherapy and targeted therapy.
  - Smoking effects on cytochrome P450 enzymes may include altered drug clearance time and plasma concentration, potentially impacting the efficacy of certain drugs. Health care professionals should consider whether patients are at risk for altered drug metabolism due to smoking and determine if medication or dose adjustments may be required. Drugs whose metabolisms are known to be affected include erlotinib, irinotecan, and bendamustine.[4-8]
- Smoking increases risk of RT-associated treatment complications during RT and may decrease treatment response.[9-11] Patients who receive RT and continue to smoke may also have an increased risk of developing a second primary cancer.[12,13,14]
- Smoking is associated with increased rates of postoperative complications and mortality after cancer surgery.
  - Compared with those who do not smoke, patients who smoke may experience decreased health-related quality of life after cancer surgery (eg, dyspnea, fatigue, pain).[15-17]
  - Smoking may impair wound healing following surgery for cancer.[18,19]
  - Increased infection rates, cardiovascular and pulmonary complications, and longer postoperative hospital stays are more commonly observed in patients who smoke.[20]
  - Survival outcomes following surgery are worse in patients who smoke.[21]

Potential Nicotine Effects on Cancer and Cardiovascular Risks (see Discussion for additional information)

- Blood nicotine levels from NRT, including combination NRT, are significantly less than from smoking cigarettes. Therefore, health care professionals and people who smoke should not be dissuaded from using NRT to foster quitting and long-term abstinence.
- There is insufficient evidence that NRT causes cancer in humans.[22-25]
- There is insufficient evidence that NRT increases the risk of myocardial infarction or cardiovascular disease.[26]

References

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SMOKING-ASSOCIATED RISKS FOR PATIENTS WITH CANCER
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SMOKING CESSATION AND TREATMENT RESOURCES FOR PATIENTS

Cancer-Related Resources for Patients[a]

American Cancer Society (ACS)
- Website: https://www.cancer.org/healthy/stay-away-from-tobacco.html
- Phone: 1.800.227.2345
- Guide to quitting smoking: https://www.cancer.org/cancer/risk-prevention/tobacco/guide-quitting-smoking.html

American Society of Clinical Oncology (ASCO)
- Website: https://www.asco.org/news-initiatives/current-initiatives/current-initiatives/cancer-care-initiatives/prevention-survivorship/tobacco-cessation-control/tools

American Lung Association
- Website: https://www.lung.org/stop-smoking
- Lung Cancer Helpline: 1.844.252.5864
- Program: Freedom from Smoking®: https://www.lung.org/stop-smoking/join-freedom-from-smoking

Smokefree.gov
- Website: https://www.smokefree.gov
- NCI Quitline: 1.877.448.7848
- LiveHelp: https://livehelp.cancer.gov/app/chat/chat_launch
- quitSTART App: https://smokefree.gov/tools-tips/quitstart
- SmokefreeText Messaging Programs (available for adults, teens, pregnant persons, and veterans, and in Spanish): https://smokefree.gov/smokefree-text-messaging-programs or text "QUIT" to 47848
- "Clearing the Air: Quit Smoking Today": https://smokefree.gov/sites/default/files/pdf/clearing-the-air-accessible.pdf

General Resources for Patients[a]

American Heart Association
- Website: https://www.heart.org/en/healthy-living

American Indian Commercial Tobacco Program
- Website: https://americanindian.quitlogix.org
- Phone: 1.855.372.0037

Asian Smokers' Quitline
- Website: https://www.asiansmokersquitline.org
- Mandarin/Cantonese: 1.800.838.8917; Korean: 1.800.556.5564; Vietnamese: 1.800.778.8440

Ex: A New Way To Think About Quitting Smoking
- Website: https://www.becomeanex.org
- ExCommunity: https://excommunity.becomeanex.org

State/Territory Quitlines and Services
- Website: https://www.naquitline.org/page/mapus

Centers for Disease Control and Prevention (CDC) Office on Smoking and Health
- Website: https://www.cdc.gov/tobacco/campaign/tips/quit-smoking/index.html
- English: 1.800.QUIT.NOW (1.800.784.8669)
- Spanish: 1.855.335.3569 (1-855-DÉJELO-YA)
- Smokefree Text Messaging Programs (state and national services available for adults): Text QUITNOW to 333888 or https://www.cdc.gov/tobacco/campaign/tips/quit-smoking/national-texting-portal.html

VA How to Quit
- Website: https://www.mentalhealth.va.gov/quit-tobacco/how-to-quit.asp

TRICARE and UCanQuit2 (Military and family)
- Website: https://www.tricare.mil/HealthWellness/Tobacco.aspx
- Text support: https://www.ycq2.org/resources/smokefreemil

Pharmacies Providing Cessation Services
- Website: https://www.QuitSmokingPharmacies.com

[a] Free or low-cost support is sometimes available for patients who are uninsured or underinsured. Resources may vary by organization and location. Contact each organization to learn more about the availability of free and low-cost smoking cessation and treatment resources offered regionally.

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SMOKING CESSATION AND TREATMENT RESOURCES FOR PATIENTS

General Resources for Patients for E-Cigarette/Vaping Cessation and Treatment[a]

Smokefree.gov
- Website: https://smokefree.gov/quit-vaping-resources
- NCI Quitline: 1.877.448.7848
- LiveHelp: https://livehelp.cancer.gov/app/chat/chat_launch
- quitSTART App: https://smokefree.gov/tools-tips/quitstart
- Smokefree Text Messaging Programs (available for adults, teens, pregnant persons, and veterans, and in Spanish): https://smokefree.gov/smokefree-text-messaging-programs or text "QUIT" to 47848.
- "Clearing the Air: Quit Smoking Today": https://smokefree.gov/sites/default/files/pdf/clearing-the-air-accessible.pdf

Truth Initiative
- Website: https://truthinitiative.org/exprogram

[a] Free or low-cost support is sometimes available for patients who are uninsured or underinsured. Resources may vary by organization and location. Contact each organization to learn more about the availability of free and low-cost smoking cessation and treatment resources offered regionally.

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SMOKING CESSATION AND TREATMENT RESOURCES FOR HEALTH CARE PROFESSIONALS

Cancer-Related Resources for Health Professionals

American Association for Cancer Research (AACR)
- Policy Statement: https://cancerres.aacrjournals.org/content/70/9/3419?ijkey=9980cf69559104118cb06580ca278aea35f5a899&keytype2=tf_ipsecsha

American Society of Clinical Oncology (ASCO)
- Tobacco Cessation Guide for Oncology Providers: https://cdn.bfldr.com/KOIHB2Q3/as/pjv87h5qvvs8jv4jms4f9g/Tobacco-Cessation-Guide

American College of Surgeons Commission on Cancer (Standard 5.9: Smoking Cessation for Patients with Cancer)
- Website: https://accreditation.facs.org/accreditationdocuments/CoC/Resources/SmokingCessation/Standard5.9.pdf

Cancer Center Cessation Initiative Tobacco Treatment Roadmap for Cancer and Other Patients
- Website: www.TobaccoTreatmentRoadmap.org

NCCN Guidelines for Lung Cancer Screening
- Website: https://www.nccn.org/professionals/physician_gls/pdf/lung_screening.pdf

NCCN Guidelines for Survivorship
- Website: https://www.nccn.org/professionals/physician_gls/pdf/survivorship.pdf

NCI- Physician Data Query (PDQ)
- Website: https://www.cancer.gov/about-cancer/causes-prevention/risk/tobacco/quit-smoking-hp-pdq

U.S. Department of Health and Human Services, Public Health Service
- Treating Tobacco Use and Dependence: 2008 Update. Content last reviewed June 2015. Agency for Healthcare Research and Quality, Rockville, MD. https://www.ahrq.gov/professionals/clinicians-providers/guidelines-recommendations/tobacco/index.html

General Resources for Health Professionals

American Academy of Family Physicians (AAFP)
- Ask and Act Tobacco Cessation Program: https://www.aafp.org/family-physician/patient-care/care-resources/tobacco-and-nicotine/ask-act.html

American Thoracic Society
- Initiating Pharmacologic Treatment in Tobacco-Dependent Adults: An Official ATS Clinical Practice Guideline Implementation Tools: https://www.thoracic.org/statements/guideline-implementation-tools/pharmacologic-treatment-of-tobacco-dependence.php

Million Hearts® 2022
- Tobacco Cessation Change Package: https://millionhearts.hhs.gov/tools-protocols/action-guides/tobacco-change-package/index.html

Rx for Change: Clinician-Assisted Tobacco Cessation
- Website: https://rxforchange.ucsf.edu

Society for Research on Nicotine & Tobacco
- Website: https://www.srnt.org

U.S. Department of Health and Human Services
- Tobacco Reports and Publications: https://www.hhs.gov/surgeongeneral/reports-and-publications/tobacco/index.html

American College of Chest Physicians (CHEST)
- Toolkit: https://tobaccodependence.chestnet.org

Association for the Treatment of Tobacco Use and Dependence (ATTUD)/Council for Tobacco Treatment Training Programs (CTTTP)
- Website: https://www.attud.org
- Accredited training programs: https://ctttp.org/accredited-programs

National Behavioral Health Network for Tobacco & Cancer Control
- Website: https://www.bhthechange.org

Smokefree.gov
- Website: https://smokefree.gov/health-care-professionals

U.S. Preventive Services Task Force (USPSTF)
- Final Recommendation Statement: Tobacco Smoking Cessation in Adults, Including Pregnant Persons: Interventions. U.S. Preventive Services Task Force. January 2021. https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/tobacco-use-in-adults-and-pregnant-women-counseling-and-interventions

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PRINCIPLES OF SMOKING CESSATION AND CANCER SURGERY

- People who smoke who need surgery should be treated with evidence-based smoking treatment approaches, including pharmacotherapy, in order to improve surgical outcomes and long-term abstinence from smoking. Initial therapy with combination NRT or varenicline is recommended. There is substantial evidence that the combination of behavioral treatment and pharmacotherapy is more effective than either one alone.
  - For primary therapy and follow-up recommendations, see SC-4. Preoperative pharmacotherapy options depend on urgency of surgery, within one week or more than one week. Options include:
    - ◊ If <7 days to surgery, combination NRT is preferred (category 1) because guidelines for quitting with NRT include a quit date within 24 hours, while varenicline is typically 7 days. However, varenicline is an option; in some clinical situations the quit date for varenicline may be sooner.
    - ◊ If ≥7 days to surgery, either combination NRT (category 1) or varenicline (category 1) are options. Smoking treatment should happen as soon as possible; ideally, the patient should be abstinent for ≥4 weeks, but this recommendation should not cause a delay in surgery.[1]
    - ◊ NRT use is not a contraindication to surgery. There is no evidence that NRT degrades the wound-healing benefits of abstinence from smoking in humans. NRT offers benefits over continued smoking. While some data from animal studies have suggested an association between nicotine and decreased wound healing, no such studies exist in humans.[2-4] NRT typically provides less nicotine than cigarettes, and nearly doubles the chance of smoking abstinence.
- Smoking increases the risk of pulmonary complications, surgical site infection, and poor wound healing in patients undergoing surgery. See Smoking-Associated Risks for Patients With Cancer (SC-B). Quitting smoking reduces these complications. There is no evidence that quitting smoking shortly before surgery increases the risks of any complications.
- Patients should be encouraged to quit smoking as soon as possible before surgery, regardless of how short the time is to surgery. Even quitting on the day of surgery has a benefit.
- Longer periods of abstinence from smoking confer better surgical outcomes but should not delay appropriate timing for cancer resection.
- For patients who need surgery, quitting smoking is recommended as soon as possible. It is not recommended to wait until after surgery.
- Although quitting smoking reduces surgical risk, access to cancer surgery should not be restricted solely on the basis of continued smoking.

[1] https://www.quitforsurgery.com.
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[3] Sørensen LT. Wound healing and infection in surgery: the pathophysiological impact of smoking, smoking cessation, and nicotine replacement therapy: a systematic review. Ann Surg 2012;255:1069-1079.
[4] Stefan MS, Pack Q, Shieh MS, et al. The association of nicotine replacement therapy with outcomes among smokers hospitalized for a major surgical procedure. Chest 2020;157:1354-1361.

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PRINCIPLES OF BEHAVIORAL TREATMENT OF SMOKING

- Pharmacotherapy is most effective when combined with behavioral treatment.[1] Brief advice is better than nothing.
  - While implementing evidence-based smoking cessation remains the responsibility of all health professionals, population-level studies of smoking treatment modalities indicate that counseling plus medication results in a significant improvement in cessation rates relative to no counseling or medication.[2] In addition to the benefits of enhancing motivation and knowledge of the addiction process, behavioral treatment assists patients with medication use and strategies since adherence to tobacco treatment medication recommendations is often inadequate. Therefore, pharmacotherapy alone without some form of counseling may not be better than quitting unaided.
- Through behavioral treatment, people who smoke learn coping skills and receive support and education. Behavioral treatment, tailored to the patient's tobacco- or nicotine-containing product use and previous quit attempts, provides strategies for:[1-3]
  - Coping with nicotine withdrawal symptoms and cravings (Note: Most nicotine withdrawal symptoms manifest in the first 1–2 days, peak within the first week, and subside within 2–4 weeks of quitting). Increased appetite, weight gain, and cravings may persist for several months.
  - Identifying what triggers them to smoke (eg, coffee, alcohol, social situations, stress).
  - Coping with stressful and difficult situations in which smoking is likely, avoiding high-risk situations.
  - Addressing other patient-specific barriers to and facilitators of smoking behavior change.
  - Coping with obstacles such as pain, a desire to maintain personal control, second-hand smoke exposure, guilt over smoking, fear of stigmatization, and fatalism regarding disease.
  - Contingency management may be considered as an adjunct behavioral treatment when standard counseling and pharmacotherapy alone are insufficient, or when rapid abstinence is clinically important (eg, prior to surgery).
- Therapy should include skills training, social support, and motivational interviewing with print- or web-based patient education materials.
- Specialized treatment centers should provide treatment for smoking targeted specifically to patients with cancer (eg, individual therapy and group support that focus on challenges specific to cancer survival and treatment). Therapy should be performed by a Tobacco Treatment Specialist or a dedicated staff member (ie, nurse, medical assistant, health educator) trained in the evidence-based treatment of smoking.
- In people with cancer who smoke, there is a high incidence of depression, anxiety, and stress, all of which are common causes of return to smoking. Health care professionals should routinely assess for these symptoms throughout treatment for smoking. It may be optimal to enroll patients in a behavioral treatment program with specific interventions designed to ameliorate these conditions and other cancer-related challenges that may increase risk for return to smoking. This may require referral for appropriate evaluation and treatment as needed or to specialized smoking treatment programs that have staff trained to treat psychiatric and/or other substance use disorders.
- Smoking by other household members is an important predictor for not remaining abstinent and exposes the patient to secondhand smoke. To benefit the patient, individuals in the household and other shared spaces should be encouraged to abstain from smoking; resources may be provided.
- Specific populations may have particular concerns to consider in treatment:
  - Increased return to smoking risk among patients living with HIV,[4] psychiatric disorders,[5] other substance use disorders,[6] those experiencing homelessness or incarceration, or those who identify as lesbian, gay, bisexual, transgender, queer, and additional sexual orientation and gender identities (LGBTQ+).
  - High rates of menthol cigarette use that may increase nicotine use in African American communities and other groups that have been historically marginalized.[7,8]
  - Preserving sacred use of tobacco while reducing commercial tobacco use in American Indian or Native American communities.[9]
- For more information on behavioral treatment for smoking cessation, see Discussion.

References

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PRINCIPLES OF BEHAVIORAL TREATMENT OF SMOKING

Behavioral and Cognitive Behavioral Treatment

- Four or more sessions are recommended during each course of pharmacotherapy, with additional sustained counseling as needed. The first session is recommended within 3 weeks of cessation (within 1 week preferred as a majority of people who smoke return to smoking in this time frame).[10]
  - Brief advice of about 3 minutes by health care professionals is associated with a small but important increase in smoking abstinence rates.[3] Longer, more frequent sessions are associated with higher abstinence rates.[11] The maximum benefit is with intensive and sustained counseling with trained counselors.[12,13]
- Individual or group therapy should take place in-person and/or by phone/telehealth,[14] in coordination with a smoking treatment clinic if available.[15,16,17]
  - For those in active cancer treatment, behavioral treatment can occur during scheduled oncology visits to avoid the need for additional appointments.
  - Refer to evidence-based resources for quitting, such as a smoking cessation quitline, in addition to providing brief advice from a health care professional, if face-to-face or group intervention is not available. See Smoking Cessation and Treatment Resources for Patients/Health Care Professionals (SC-C).
    - ◊ Quitlines are a source of brief behavioral treatment for quitting smoking and may have added value, especially in lower-resource settings.

Motivational Counseling

- Motivational counseling is beneficial for all patients throughout treatment but is essential for those not immediately ready to quit.[3,18,19]
  - Motivational counseling involves first establishing a connection with the patient and then exploring feelings (eg, distress), beliefs (eg, fatalistic), and values (eg, importance of family) in order to help the patient address ambivalence or conflicting motivations. Clinicians use a guiding and empathetic style to help the patient: 1) understand their motivations; 2) understand the relation of these to their values and goals; and 3) build their confidence to achieve and maintain abstinence. To achieve this, clinicians ask open-ended questions, reflect patient emotions and concerns, affirm patient strengths, and summarize important points. Clinicians should express acceptance for the patient and validate the patient's worth, strengths, and independence.
  - Physicians and health professionals use four basic strategies, commonly referred to as OARS, for motivational interviewing: Open-ended questions; Affirmations; Reflective listening; and Summary statements.
  - For patients who struggle to identify reasons to quit, the clinician can also use the 5 R's strategy focusing on: the personal Relevance of quitting; personal Risks of continuing smoking; personal Rewards of quitting; identifying Roadblocks to quitting; and Repeating the message at every contact.[3]

Contingency Management

- Contingency management is the use of structured financial incentives contingent on biochemically verified abstinence or engagement. It is a behavioral treatment to increase smoking cessation rates.[20,21]
- Contingency management may be particularly helpful for patients with cancer at elevated risk for continued smoking, including those experiencing high stress, depression or anxiety, financial strain, or other social determinants that interfere with treatment engagement.
- Contingency management should be delivered using objective biochemical verification, typically expired carbon monoxide (CO) or equivalent cotinine testing. Verification frequency should be highest during the first several weeks after cessation, when risk of return to smoking is greatest.
- Incentives should be structured, transparent, and time limited.
- Contingency management is most effective when integrated with standard counseling and pharmacotherapy.
- Programs should specify the target behavior, verification method, schedule of reinforcement, and duration of contingency management.

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PRINCIPLES OF BEHAVIORAL TREATMENT OF SMOKING
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[4] Vijayaraghavan M, Penko J, Vittinghoff E, et al. Smoking behaviors in a community-based cohort of HIV-infected indigent adults. AIDS Behav 2014;18:535-543.
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PRINCIPLES OF PHARMACOTHERAPY FOR SMOKING

- Abstinence from smoking improves clinical outcomes for patients with cancer. Therefore, the included agents and methodologies should be made available to all patients with cancer who smoke.
- A minimum of 12 weeks of combination NRT or varenicline[a] is recommended for the initial quit attempt. Therapy may be extended to ≥6 months, when clinically indicated, to promote continued abstinence.
  - Follow-up is recommended (in-person or by phone/telehealth) within 3 weeks (within 1 week preferred)[1] after starting pharmacotherapy, but can be adjusted to coordinate with regularly scheduled oncology appointments as needed. Additional periodic follow-up during therapy (at a minimum of 12-week intervals), and after completion of therapy, is recommended.
  - Most nicotine withdrawal symptoms manifest in the first 1–2 days, peak within the first week, and subside within 2–4 weeks of quitting. Increased appetite, weight gain, cravings, and mood changes may persist for several months. It is important to distinguish between withdrawal symptoms from treatment-related side effects in order to not discontinue therapy early. Encourage continued therapy through brief slips. Patients who do not quit immediately may quit at some later point after withdrawal symptoms subside.
  - Health care professionals should educate patients to take smoking treatment medications according to the prescribed regimen and for the full course of therapy.
  - Pharmacotherapy dose adjustments, including increasing the dosages of NRT and varenicline (but not bupropion), may be considered. In some cases, dose reduction is warranted based on clinical indicators.
- Track attempts at smoking reduction. If reduction efforts plateau, or reaching total abstinence seems unlikely, consider switching to a different pharmacotherapy.
- As patients progress through multiple lines of treatment, behavioral treatment should be progressively intensified with referral to specialty care (eg, psychiatrist, psychologist) as indicated.
- For those being treated with varenicline, the addition of NRT or bupropion can be considered as clinically indicated to maintain abstinence despite continued urges to smoke. Short-acting NRT can be added to varenicline to control/manage cravings. For patients who are heavily dependent and/or have unsuccessfully tried varenicline, varenicline + patch can be considered.
- When considering pharmacotherapy options, consider cultural and contextual factors including cost, ability to pay, insurance coverage, ability to adhere to the treatment regimen, and patient preferences. Address patient concerns about pharmacotherapy benefits, myths, and risks.

Pharmacotherapy Options[a]	
Preferred	**Other Recommended**
• Combination NRT:[b] Long-acting NRT (nicotine patch) + short-acting NRT (lozenge/gum/nasal spray) • Varenicline[c] or • Switch to alternate preferred primary therapy option not previously used (combination NRT or Varenicline)[a-c]	• Varenicline[c]/short-acting NRT • Bupropion[d] (eg, for patients with depression or fatigue)[a] • Bupropion[d]/NRT[b] (eg, for patients with depression or fatigue)[a] • Bupropion[d]/Varenicline[c] • Varenicline[c]/nicotine patch (category 2B)

[a] Decision to continue or switch therapy should be based on prior cessation success, patient preference, toxicity, and/or a change in clinical status (eg, upcoming surgery).

[b] Any form of NRT or combination NRT (nicotine patch + short-acting NRT) may be used. Studies evaluating the efficacy of varenicline combined with NRT utilized the nicotine patch (Koegelenberg CF, et al. JAMA 2014;312:155-161).

[c] Nausea is a common side effect of varenicline and may need to be managed for patients with cancer, especially during chemotherapy. Varenicline should be used with caution for those with seizure risk.

[d] Bupropion should be avoided in patients with seizure risk, including those with brain metastases or primary brain tumors.

References

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PRINCIPLES OF PHARMACOTHERAPY FOR SMOKING

	Standard Dose/Administration[e,f,g]	Duration
Combination NRT (preferred)	• Initiate dosing 1–2 weeks prior to agreed upon quit date, while patient is actively cutting down cigarettes per day • Begin with nicotine patch + short-acting NRT ▸ If smoking 1–10 cigarettes/day, use 14-mg patch ▸ If smoking >10 cigarettes/day, use 21-mg patch ◊ If initial patch is not effective, consider increasing the dose, including up to more than one patch • Short-acting gum or lozenge: 2 or 4 mg ▸ 2 mg preferred if time to first cigarette is >30 minutes after waking; or 4 mg preferred if time to first cigarette is ≤30 minutes after waking ▸ Every 1–2 h (while awake and as tolerated), or more often as needed • Short-acting nasal spray ▸ 1 spray (0.5 mg) in each nostril ▸ Start with 1 or 2 doses per hour, may be increased up to a maximum dose of 40 mg (80 sprays) per day	Minimum of 3 months (6 months or longer preferred) of pharmacotherapy is recommended.[3-9] However, therapy may be substantially extended to promote continued abstinence
Varenicline (preferred)	• Initiate dosing 1–2 weeks prior to agreed upon quit date, while patient is actively cutting down cigarettes per day • Days 1–3: 0.5 mg orally, once daily • Days 4–7: 0.5 mg orally, twice daily • Day 8 to end of treatment: 1 mg orally, twice daily (if tolerated) ▸ Consider increase to 3 mg per day (1.5 mg twice daily if tolerated) for those who cut back by ≥50% but have not quit at 6 wks ▸ If severe renal impairment (estimated creatinine clearance <30 mL/min): Begin with 0.5 mg once daily and titrate to 0.5 mg twice daily ▸ For patients with end-stage renal disease undergoing hemodialysis, 0.5 mg maximum daily, if tolerated	
Bupropion	• Initiate dosing 1–2 wks prior to agreed upon quit date, while patient is actively cutting down cigarettes[2] • Days 1–3: 150 mg orally, once daily • Day 4 through end of treatment: ▸ Sustained release: 150 mg orally, twice daily, if tolerated; or ▸ Extended release: 300 mg, once daily, if tolerated • Maximum 300 mg per day • Adjust dose or frequency for: ▸ Renal impairment[g] ▸ Hepatic impairment: Maximum dose 150 mg every other day for moderate/severe hepatic impairment (Child-Pugh score 7–15); for mild hepatic impairment (Child-Pugh score 5–6), consider reducing the dose and/or frequency adjustment	

[e] Dose adjustments may be considered if clinically indicated.
[f] See drug labels and full prescribing information for varenicline, bupropion, and NRT products.
[g] Gradually decrease dose over 10 weeks or more. Dose reduction may not be appropriate for patients with limited life expectancy.

Adverse Effects and Contraindications (SC-F 3 of 4)

References

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PRINCIPLES OF PHARMACOTHERAPY FOR SMOKING

- In most circumstances, the side effects related to preferred smoking treatment medications are minimal and are considered an acceptable risk compared to smoking. Serious side effects are extremely rare. Refer to manufacturer inserts for exhaustive lists of potential side effects and warnings.
- A multicenter randomized controlled trial (RCT) examined the neuropsychiatric safety of varenicline and bupropion sustained release in 2 cohorts of patients: those with diagnosed psychiatric disorders (n = 4074) and those without (n = 3984). Rates of neuropsychiatric adverse events in individuals receiving varenicline or bupropion sustained release were not significantly increased relative to those receiving nicotine patches or placebo in either cohort.[g,10]

	Adverse Effects and Contraindications[f,10]
Combination NRT (preferred)	Blood nicotine levels from NRT, including combination NRT, are significantly less than from smoking cigarettes. NRT is well tolerated and nicotine toxicity is rare and transient, even when used with smoking. Patients commonly underdose when using combination NRT. Significant nicotine toxicity is rare, but possible and usually short-lived.
Varenicline (preferred)	• Nausea is a common side effect of varenicline and may need to be managed for patients with cancer, especially during chemotherapy. ▸ Take varenicline after eating and with a full glass of water to minimize nausea risk. • Although the literature does not support that it increases the risk for neuropsychiatric issues,[10] health care professionals should monitor for the development or worsening of serious neuropsychiatric issues (ie, depression and suicidal ideation/behavior), including for patients without a previous history, and discontinue use if these signs occur. • Use with caution in patients with a history of seizure disorder or at risk for seizures. • For patients experiencing nicotine withdrawal while on varenicline, NRT may be considered.
Bupropion	• Although these side effects are uncommon, health care professionals should monitor for the development or worsening of serious neuropsychiatric issues (ie, depression and suicidal ideation/behavior), including for patients without a previous history, and discontinue use if these signs occur. • Bupropion should be avoided in patients with seizure risk, including those with brain metastases or primary brain tumors. • Bupropion is contraindicated for patients with seizure disorder, those taking monoamine oxidase inhibitors (MAOIs) (increased risk of hypertensive reactions), those with current diagnosis or history of anorexia or bulimia, or those who have abrupt discontinuation of alcohol, benzodiazepines, barbiturates, or antiepileptic drugs. • Inhibitors of CYP2B6 (eg, ticlopidine, clopidogrel) can increase bupropion (active drug) and require dose adjustments. Consider dose reduction when using with bupropion. Use bupropion with caution when used with drugs that lower seizure threshold. Caution should be used for bupropion for patients treated with tamoxifen. Central nervous system (CNS) toxicity can occur when bupropion is used concomitantly with dopaminergic drugs (eg, amantadine, levodopa).

[f] See drug labels and full prescribing information for varenicline, bupropion, and NRT products.

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References

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ABBREVIATIONS

CDER	**Center for Drug Evaluation and Research**
CNS	**central nervous system**
EVALI	**e-cigarette or vaping product use-associated lung injury**
LGBTQ+	**lesbian, gay, bisexual, transgender, queer, and additional sexual orientation and gender identities**
MAOI	**monoamine oxidase inhibitor**
MRTP	**modified risk tobacco product**
NRT	**nicotine replacement therapy**
RCT	**randomized controlled trial**

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NCCN Categories of Evidence and Consensus	
Category 1	Based upon high-level evidence (≥1 randomized phase 3 trials or high-quality, robust meta-analyses), there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2A	Based upon lower-level evidence, there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2B	Based upon lower-level evidence, there is NCCN consensus (≥50%, but <85% support of the Panel) that the intervention is appropriate.
Category 3	Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

All recommendations are category 2A unless otherwise indicated.

NCCN Categories of Preference	
Preferred	Interventions that are based on superior efficacy, safety, and evidence; and, when appropriate, affordability.
Other recommended	Other interventions that may be somewhat less efficacious, more toxic, or based on less mature data; or significantly less affordable for similar outcomes.
Useful in certain circumstances	Other interventions that may be used for selected patient populations (defined with recommendation).

All recommendations are considered appropriate.

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National Comprehensive Cancer Network®

NCCN Guidelines Version 1.2026
Smoking Cessation

Discussion

This discussion corresponds to the NCCN Guidelines for Smoking Cessation. Last updated: December 3, 2025.

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Overview

Tobacco smoking has been implicated in causing cancers of the lungs, mouth, lips, nose, sinuses, larynx, pharynx, esophagus, stomach, pancreas, kidney, bladder, uterus, cervix, colon/rectum, and ovary, and myeloid leukemia.[1] Cancers linked to tobacco use comprise 40% of all cancer diagnoses, and cigarette smoking is linked to 30% of all cancer-related deaths nationwide.[2] State-level data suggest that cigarette smoking is responsible for as high as 40% of cancer-related deaths in some geographic regions.[3] Lung cancer is the leading cause of cancer-related death in both males and females.[4]

Smoking tobacco after a cancer diagnosis decreases the efficacy of cancer treatments, increases the side effects of cancer treatment, increases the risk of death from all causes, and adds to treatment complexities and the cost of cancer care, as well as the probability of new or worsening comorbidities.[5-8] Smoking cessation at any time after a cancer diagnosis is associated with improved cancer outcomes, enhanced quality of life, and reduced disease among cancer survivors.[9-14] Yet, 29.2% of cancer survivors aged 18 to 44 years smoke cigarettes, which is double the national prevalence of smoking in the United States.[9]

A life-threatening cancer diagnosis should be a strong incentive to quit smoking; however, cancer survivors face unique challenges. Nonetheless, the benefits of cessation are exceptionally compelling for cancer survivors. For these reasons, the NCCN has developed these guidelines to emphasize the importance of smoking cessation in all patients with cancer and seek to establish evidence-based recommendations tailored to the unique needs and concerns of patients with cancer. The recommendations contained herein describe interventions for cessation of all combustible tobacco products (eg, cigarettes, cigars, hookah), including smokeless tobacco products. However, recommendations are based on studies of cigarette smoking.

Smoking Prevalence in Patients with Cancer

Smoking often persists beyond cancer diagnosis and treatment and well into survivorship. Results from the 2020 National Health Interview Survey showed that 12.2% of patients with cancer reported that they currently smoke.[15] Results from this survey also showed that patients with cancer who are age ≥65 years are less likely to report current smoking (7.4%), compared to patients who are aged 18 to 44 years (21.3%). A recent study of 4326 participants who were enrolled in SWOG cancer clinical treatment trials from 2016 to 2022 also showed that about12% currently smoke.[16] Additionally, this study pointed out that smoking was most prevalent among male patients aged ≥65 years and patients from vulnerable socioeconomic and demographic populations.

In the Cancer Care Outcomes Research and Surveillance (CanCORS) cohort of patients with lung and colorectal cancers, at time of diagnosis, 39% of those with lung and 14% of those with colorectal cancer reported current smoking, of which 14% with lung and 9% with colorectal cancer continued to smoke at 5 months post-diagnosis.[17] Another study revealed smoking prevalence to be highest among survivors of bladder, lung, and ovarian cancers.[18] Finally, other studies have found a high prevalence of continued smoking in survivors of cervical cancers.[19]

Health Care Community Response

Major cancer organizations have identified the need to treat tobacco use as aggressively as treating cancer in patient care.[20,21] More than 80% of oncology care-providing hospitals offer smoking cessation programs, with nationally designated cancer centers being more likely to offer programs.[22] However, despite general consensus on the importance of smoking cessation, particularly for patients with cancer, many cancer centers and

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oncology practices report that they fall short of providing consistent, high-quality smoking cessation services. In a survey of 58 National Cancer Institute (NCI)-designated cancer centers, 20% reported offering no smoking cessation services for their patients, 38% did not routinely provide tobacco educational materials to patients, and 53% reported that they effectively identified tobacco use in their patients.[23] The AACR Task Force on Tobacco and Cancer found that few cancer care institutions utilize systematic and consistent mechanisms to foster cessation among patients with cancer.[20]

Several studies have linked increased patient satisfaction to the delivery of smoking cessation advice or intervention.[24,25] Data from large surveys of oncologists practicing in academic medical centers, non-academic hospitals, and oncology practices depict generally high rates of smoking assessment and providing initial advice to quit.[26-29] However, smoking assessment rates were weaker outside of the academic/university setting (ie, for those practicing in a hospital-based, non-academic, or private setting).[29] Results from a survey of 28 NCI-designated cancer centers showed that the proportion of patients who smoke who receive at least some evidence-based smoking treatment (ie, counseling and/or pharmacotherapy) can be improved by whole-center involvement and increased staff-to-patient ratio.[30] However, increasing staff may not always be a viable option. Smoking cessation effectiveness was associated with electronic health record (EHR) referrals delivered via a closed-loop system (ie, patient successfully referred as appropriate, with documentation).

Regardless of work setting, only 30% to 44% of respondents reported discussing specific interventions or providing subsequent follow-up. Moreover, the majority of respondents reported inadequate training and/or a lack of confidence in ability to provide effective smoking cessation counseling and intervention.[26-29] A dearth of smoking assessment and documentation has also been demonstrated in oncology trials.[31,32]

Issues regarding insurance coverage and provider reimbursement for smoking cessation assessment, counseling, and cessation aids have also presented a challenge for the oncology community in the past. However, implementation of the Affordable Care Act has led to changes designed to increase access to smoking cessation interventions.[33] An intervention study conducted at an NCCN Member Institution showed that use of EHR functionality can be used to ensure that tobacco use is assessed and addressed in health care settings.[34] A retrospective study from an NCCN member institution showed that standardized tobacco use assessment with opt-out referral to smoking treatment was feasible and associated with increased treatment referral and acceptance rates.[35] Specifically, patients are automatically referred for services and therefore will need to proactively decline the services. A model such as this may reduce clinician bias and increase equitable access to tobacco treatment.

The NCI states that smoking cessation treatment can be incorporated into every step of the cancer care continuum (screening, diagnosis, treatment, and survivorship).[15] They have identified the following strategies to support the dissemination, adoption, and reach of smoking cessation treatment programs in cancer care clinical settings:

- “Establish an evidence-based standard of smoking cessation care across cancer clinical delivery systems that includes tobacco user identification, advice to quit, provision of or referral to evidence-based tobacco treatment, and patient follow-up.
- Measure and report the delivery of smoking cessation treatment as performance metrics for clinicians, hospitals, and health care system leadership.
- Emphasize the delivery of smoking cessation treatment as an important evaluation criterion for oncologists and cancer clinics by professional oncology organizations.

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- Implement changes in health care systems, such as using electronic health record tools and other workflow adaptations that facilitate the consistent delivery of smoking cessation interventions in cancer care.
- Develop resources that enable universal implementation of smoking cessation treatment programs in cancer care settings, including strategies that: reduce clinician burden; enhance clinical workflow integration; and provide patients with easy access to multiple treatment options."

Smoking Cessation in Patients with Cancer

Given the complexity of smoking cessation interventions for patients with cancer, there is a great need for resources that provide guidance on smoking cessation, specifically for this patient population. The NCCN Clinical Practice Guidelines (NCCN Guidelines®) for Smoking Cessation have been created to establish recommendations for smoking cessation in patients with cancer. The NCCN Guidelines Panel has developed these guidelines in order to facilitate implementation of this standard, to allow for quality control monitoring, to fill a gap among existing treatment guidelines, and ultimately, to improve the health and outcomes for patients with cancer.

Smoking cessation in this patient population presents a number of challenges. Patients may have limited awareness about the harms associated with continued smoking upon a diagnosis of cancer. A survey assessed knowledge of the harms of continued smoking in 1118 patients who currently or formerly smoked and had a cancer diagnosis.[36] Across the board, 38% to 62% of the respondents were unaware of the negative impact that smoking can have on chemotherapy, radiation therapy (RT), surgery, overall treatment efficacy, survival, and development of second primary cancers. Lack of awareness was associated with current smoking at time of diagnosis.

Barriers in Smoking Cessation

In the general population, individuals who smoke report a number of different barriers to quitting, including stress; dependence; home, work, and social environmental factors; and lack of resources and support for quitting.[37] Importantly, patients and health care professionals in the oncology setting face additional life challenges that can amplify the magnitude of these barriers. However, pain, desire to maintain personal control, secondhand smoke exposure, guilt over smoking, fear of stigmatization, and fatalism regarding disease also represent obstacles unique to oncology patients, particularly those with advanced disease.[38-44] A study that examined whether cancer-related disease factors adversely affected engagement with smoking cessation treatment with varenicline and behavior counseling showed that disease factors such as tumor site, cancer treatment, and time since diagnosis were not associated with medication adherence, counseling adherence, or abstinence.[45] Although health-related quality of life (HRQOL) was predictive of medication adherence, abstinence rates were not impacted. Use of multiple tobacco products is associated with more smoking and greater nicotine dependence.[46,47]

In a population-based analysis of individuals recently diagnosed with cancer who actively smoked, health professional-provided cessation counseling was provided to only 52% of individuals over the course of 12 months.[48] Surveys of oncology providers have identified common themes among barriers to smoking cessation for patients with cancer. In a large survey, medical and radiation oncologists reported a preference for smoking cessation interventions to be managed by other members of the health care team.[49] Inadequate provider training, lack of time, and perceived patient harm/benefit are often cited by oncology providers as

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barriers to successful intervention.[27,50] Providers have also cited patient-related factors such as inability to quit, lack of motivation, or resistance to treatment.[26,27] Several groups have described interventions to improve referral to smoking cessation services for patients with cancer by successfully integrating smoking cessation treatment into the cancer care plan.[51-54]

Alcohol use is a barrier to successful smoking treatment. A survey study including 923 adults who smoke cigarettes and who had participated in a smoking cessation intervention showed that both moderate (adjusted odds ratio [OR], 1.54; 95% CI, 1.05–2.23) and heavy drinking (adjusted OR, 2.39; 95% CI, 1.33–4.28) were associated with continued smoking at 1-month posttreatment, while only heavy drinking was associated with continued smoking at 7 months posttreatment (adjusted OR, 2.32; 95% CI, 1.35–3.96).[55] Alcohol use, as well as other substances such as cannabis, is discouraged for those attempting to quit smoking.

Notably, clinical trial research on smoking cessation for patients with cancer is limited, particularly for patients thought to have non-tobacco–related cancers. Barriers that limit or prevent enrollment in smoking cessation trials include smoking rate, medical history, contraindicated medications, lack of interest, and language barriers.[56] A study describing outcomes from the tobacco treatment program at MD Anderson Cancer Center reported 3-, 6-, and 9-month self-reported abstinence rates of 45.1%, 45.8%, and 43.7%, respectively.[57] One randomized controlled trial (RCT) compared long-term smoking cessation outcomes of patients with cervical cancer undergoing active cancer treatment versus those who completed treatment.[58] Results showed that those receiving active treatment were less likely to stay abstinent at 12 months. Ongoing RCTs are still testing for smoking cessation intervention plans in patients with cancer (eg, NCT02816697).

Guidelines Update Methodology

The complete details of the Development and Update of the NCCN Guidelines are available at www.NCCN.org.

Literature Search Criteria and Guidelines Update Methodology

Prior to the update of this version of the NCCN Guidelines® for Smoking Cessation, an electronic search of the PubMed database was performed to obtain key literature in the field of smoking cessation published since the previous Guidelines update, using the following search terms: ("Smoking Cessation") OR "Tobacco") AND "Neoplasms". The PubMed database was chosen because it remains the most widely used resource for medical literature and indexes peer-reviewed biomedical literature.

The search results were narrowed by selecting studies in humans published in English. Results were confined to the following article types: Clinical Trial, Phase III; Clinical Trial, Phase IV; Guideline; Practice Guidelines; Randomized Controlled Trial; Meta-Analysis; Systematic Reviews; and Validation Studies. The data from key PubMed articles as well as articles from additional sources deemed as relevant to these guidelines as discussed by the Panel during the Guidelines update have been included in this version of the Discussion section. Recommendations for which high-level evidence is lacking are based on the Panel's review of lower-level evidence and expert opinion.

Sensitive/Inclusive Language Usage

NCCN Guidelines strive to use language that advances the goals of equity, inclusion, and representation. NCCN Guidelines endeavor to use language that is person-first; not stigmatizing; anti-racist, anti-classist, anti-misogynist, anti-ageist, anti-ableist, and anti-weight-biased; and inclusive of individuals of all sexual orientations and gender identities. NCCN Guidelines incorporate non-gendered language, instead focusing on

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organ-specific recommendations. This language is both more accurate and more inclusive and can help fully address the needs of individuals of all sexual orientations and gender identities. NCCN Guidelines will continue to use the terms men, women, female, and male when citing statistics, recommendations, or data from organizations or sources that do not use inclusive terms. Most studies do not report how sex and gender data are collected and use these terms interchangeably or inconsistently. If sources do not differentiate gender from sex assigned at birth or organs present, the information is presumed to predominantly represent cisgender individuals. NCCN encourages researchers to collect more specific data in future studies and organizations to use more inclusive and accurate language in their future analyses.

General Principles of the Smoking Cessation Guidelines

The NCCN Guidelines for Smoking Cessation can be applied to any tobacco product, including cigarettes, smokeless tobacco, and vaping. As a generalization, the principles for treating tobacco use disorder of any tobacco products are similar. All patients, regardless of stage or treatment modalities, should be encouraged to achieve and maintain abstinence from all combustible tobacco products (eg, cigarettes, pipes, cigars, hookah, cigarillos), as well as electronic cigarettes (e-cigarettes)/vaping, and smokeless tobacco products. Smoking cessation has health benefits even after a cancer diagnosis, regardless of site, stage, or prognosis—namely improvement in cancer treatment outcomes, primary cancer recurrence, and secondary cancers. Importantly, a diagnosis of cancer may present a teachable moment and valuable opportunity for health care professionals to encourage smoking cessation.[45,59-61] It is the view of the NCCN Guidelines Panel that it is never too late for patients with cancer at any stage to stop smoking and experience health benefits.

The Panel recommends a multimodal approach to cessation therapy. The Panel recommends that treatment plans for all patients with cancer who smoke include the following three concurrent methods: 1) evidence-based motivational strategies and behavior therapy (counseling), which can be brief; 2) evidence-based pharmacotherapy; and 3) close follow-up with retreatment as needed.

The Panel asserts that a smoking cessation approach combining pharmacologic therapy and behavior therapy is the most effective and leads to the best results for achieving and maintaining abstinence from smoking, or smoking reduction for those not ready to quit. The two most effective pharmacotherapies are combination nicotine replacement therapy (NRT) (combined long- and short-acting NRT) and varenicline. There is a dose-response relationship for the success of counseling; high-intensity behavior therapy with multiple counseling sessions is most effective, but at least a minimum of brief counseling is needed (ie, about 3 minutes).[62-65] Quitlines are a source of brief behavioral counseling for quitting smoking and may have added value, especially in lower-resource settings.[66,67]

The Panel also emphasizes the importance of documenting smoking status and treatment plans in the patient health record. Patient health records should be updated at regular intervals to indicate changes in smoking status, quit attempts made, and interventions utilized. All patients undergoing smoking treatment should be considered for lung cancer screening (see the NCCN Guidelines for Lung Cancer Screening; available at www.NCCN.org).

The Panel emphasizes that smoking relapse and brief slips are common and can be managed. Health care professionals, including tobacco treatment specialists, should discuss this with patients and provide guidance and support to encourage continued abstinence from smoking. Additionally, health care professionals should be aware that smoking slips do not necessarily indicate a need for an alternative intervention. More than one quit attempt with the same therapy may be necessary to achieve

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long-term cessation. Health care professionals should educate patients to take smoking treatment medications according to the prescribed regimen and for the full course of therapy.

Treatment for smoking is an integral part of oncology treatment and should be continued throughout the entire oncology care continuum, from treatment, to survivorship, to end-of-life care. Emphasis on patient preferences and values is important when considering the best approach to smoking treatment in that setting. Cultural context is also important. For example, American Indians and Alaska Natives may use traditional tobacco for ceremonial or medicinal purposes.[68] Smoking cessation efforts in this population should target reduction of commercial tobacco use, not use of sacred tobacco.

E-cigarettes are not U.S. Food and Drug Administration (FDA)-approved for the treatment of smoking. Patients should be counseled toward the use of evidence-based smoking treatment approaches. For patients who choose to use e-cigarettes exclusively for smoking cessation, despite recommendations to use evidence-based pharmacotherapies, encourage behavioral counseling and abstinence from smoking, working with the patient as they continue to use e-cigarettes. As the patient becomes more confident about quitting smoking, encourage cessation of e-cigarettes, but not at the risk of relapse to smoking combustible products. For a full discussion, see the section on *Electronic Cigarettes ("E-Cigarettes")/Vaping*, below.

Smoking-Associated Risks for Patients with Cancer

Exposing cancer cells to cigarette smoke has been shown to promote a more malignant phenotype through its effects on angiogenesis and cell proliferation, migration, invasion, and survival. For a review of the preclinical data, see Sobus and Warren (2014).[69] These data have been corroborated by clinical studies. The 2020 Surgeon General's Report[70] confirms that smoking cessation reduces cumulative exposure to tobacco smoke across the life course and therefore supports the conclusion that smoking cessation reduces the risk of cancers that have been causally linked to cigarette smoking. Additionally, existing evidence is suggestive of a link between smoking and increased risk of cancer recurrence, poor treatment response, and increased treatment-related toxicity.

NCCN recommends that health care professionals should inform patients of the potential benefits of abstinence from smoking, including improved survival, treatment outcomes, and HRQOL, as well as decreased treatment-related toxicity, drug side effects, and surgical complications. Patients should receive education on the specific risks of smoking during treatment for their particular cancer and should be encouraged to abstain from smoking as far in advance as possible before initiating cancer treatment. To benefit the patient, individuals in the household should be encouraged to abstain from smoking. Prior to initiating treatment, when making decisions regarding treatment selection, dosage, and timing of initiation, health care professionals should consider patient smoking status and potential smoking-related effects.

Overall Survival and Mortality

Smoking has been linked not only to the development of disease in tobacco-related cancers, but also to prognosis upon diagnosis and risk of death during treatment. Evidence suggests that current smoking increases risk of death and negatively impacts survival for patients with cancer in a variety of disease sites, including bladder,[71-75] breast,[6,76-78] cervix,[79,80] colon/rectum,[81-84] endometrium,[85] esophagus,[86,87] head and neck,[88-92] kidney,[93-95] lung,[5,10,11,84,96,97] thyroid,[92] ovary,[98] pancreas,[99,100] and prostate,[92,101,102] as well as hematologic malignancies,[103] melanoma,[104] and brain metastases.[105]

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A number of studies of smoking at cancer diagnosis offer insight into the negative effects of smoking on overall survival (OS), disease-specific mortality (DSM), and all-cause mortality (ACM). A population-based, cross-sectional study using data from the National Health and Nutrition Examination Survey found that in 255,100 person-years of follow-up, the risk of ACM increases according to the number of cigarettes smoked per day (non-daily, <20 cigarettes per day, 20–40 cigarettes per day, and ≥40 cigarettes per day).[106] In the 2020 Surgeon General's Report, the findings of 10,975 patients from 10 studies (7 prospective cohort and 3 retrospective cohort studies) were analyzed.[70] Three prospective cohort studies compared continued smoking and smoking cessation to never smoking. Continued smoking after a cancer diagnosis significantly increased the risk of mortality compared to never smoking. The risk of ACM for those who continued to smoke was greater than for those who quit.[6,91,107] An additional three studies reported that compared to persistent smoking, quitting was significantly associated with reduced ACM in patients with either non-small cell lung cancer (NSCLC) or small cell lung cancer.[108-110] Lastly, four studies similarly reported increased ACM in disease sites including lung, colorectal, bladder, and other cancers with continued smoking relative to quitting.[5,12,107,111] The 2020 Surgeon General's Report therefore concluded that smoking cessation after a cancer diagnosis can significantly reduce ACM relative to continued smoking.[70] The quality and quantity of evidence presented was consistent with knowledge of a reduction in ACM with smoking cessation in the general population.

Risk of Recurrence or Secondary Primary Tumor

A number of studies have linked cigarette smoking and heightened risk of recurrence (ie, recurrent cancer in the same anatomic location as the original primary cancer). The 2014 Surgeon General's Report identified a positive association between smoking and risk of recurrence in 82% of the reviewed studies (42/51), with 53% of studies revealing significantly increased risk.[7] Among the studies that compared relative risk (RR) of recurrence between patients who never smoked, patients who formerly smoked, and patients who currently smoke, the median RR was 1.42 and 1.15 for current and former smoking, respectively.[7]

Disease sites with data linking current patient smoking to increased risk of recurrence include the anus,[112] bladder,[71,113,114] breast,[115] lung,[10,116] stomach,[117] and prostate.[101,102,118-121]

Studies have also examined the impact of continued smoking in patients with cancer on the risk of second primary tumor formation. The 2014 Surgeon General's Report identified a positive association between smoking and risk of second primary tumor in all studies examined (n = 26). The association was strongest when considering the effects of smoking on RR of developing a smoking-related second primary cancer (eg, lung cancer). Among five studies classifying smoking status into "never," "former," and "current," the median elevated RR of a second primary tumor was 1.20 and 2.20 for former and current smoking, respectively. Data from a large retrospective series and pooled data analyses have continued to provide support for smoking and increased risk of second primary malignancy, especially smoking-associated cancers.[122,123] Additionally, data also suggest that smoking interacts synergistically with RT to elevate the risk of second primary cancers.[7,124,125]

Smoking cessation is linked to reduced risk of recurrence and second primary tumor formation. Data from patients with lung and head and neck cancers showed that rates of second primary cancers were lower for patients who quit smoking than for those who continued to smoke after diagnosis.[126-128] In a cohort of patients with colon cancer, current smoking was associated with significantly greater risk for baseline metastasis, but interestingly, rates among those who formerly smoked and never smoked were similar and significantly less in comparison.[129]

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Smoking-Related Effects on Treatment Efficacy, Side Effects, and Outcomes

A majority of the existing data establishes and supports the detrimental impact of persistent smoking during cancer treatment. In a 2014 report from the Surgeon General, 80% of the evaluated studies (66/82) demonstrated a statistically significant association between smoking and increased anticancer treatment-related toxicity.[7] Smoking has implications across the spectrum of cancer treatment, including surgical outcomes, RT efficacy and toxicity, chemotherapy metabolism and side effects, and overall symptom burden. This Discussion also addresses the developing evidence base for the benefits of abstinence from smoking after receiving a cancer diagnosis.

Smoking-Associated Risks

Surgery

Smoking negatively impacts outcomes from cancer surgery, affecting postoperative complications, quality of life, length of hospital stay, and mortality risk.

The adverse effects of smoking on postoperative outcomes were examined in more than 20,000 patients with gastrointestinal (n = 12,432), lung (n = 4490), and urinary tract cancers (n = 3491) using the Veteran's Health Administration Surgical Quality Improvement Program (VASQIP) database from 2002 through 2008.[130] Surgical complications examined included surgical site infections, vascular complications (ie, venous thromboembolism, stroke/cerebrovascular accident, myocardial infarction), and composite pulmonary outcomes (CPO: pneumonia, failure to wean from ventilator >48 hours, or re-intubation for cardio-respiratory failure). Compared with patients who formerly smoked, patients in the gastrointestinal cancer cohort who smoke had higher postoperative rates of pneumonia, failure to wean from ventilator, reintubation, and CPO. In the lung cancer cohort, patients who smoke had higher rates of pneumonia, failure to wean from the ventilator, reintubation, CPO, and return to surgery compared with patients who formerly smoked. Current smoking status was associated with an increased length of hospital stay across all cancer sites when compared with patients who never smoked; patients who never smoked and patients who formerly smoked did not differ on this measure.

Similar findings have been reported by other groups for patients who had surgery for lung cancer, bladder cancer, and gastric cancer, and in patients with leukemia undergoing hematopoietic cell transplantation (HCT).[131-138]

Postsurgical outcomes (ie, incisional infections, infectious and major complications, mortality at 30 days) were compared between cohorts of patients who never smoked, formerly smoked, and currently smoke using data from more than 26,000 patients with colorectal cancer in the American College of Surgeons National Surgical Quality Improvement Program database (2005–2010). Postoperative morbidity and mortality rates were higher among patients who smoke, and a significant dose-dependent effect was observed when stratifying risk of major complications by pack-years of smoking.[139] Similarly, a retrospective analysis of 5715 patients from the Veterans Affairs Surgical Quality Improvement Program database found that patients who smoke had a 48% higher likelihood of suffering a pulmonary complication (95% CI, 1.03–2.14; P = .035) and 72% higher likelihood of suffering multiple complications (95% CI, 1.07–2.76; P = .035), compared to patients who recently quit smoking.[140] For patients aged >70 years, a retrospective analysis of the American College of Surgeons National Surgical Quality Improvement Program database from 2011 to 2017 found that the risk of a major complication increased from 17.8% in patients who don't smoke to 21.7% in patients who smoke (P < .001).[141]

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Smoking can also impair wound healing and predispose patients to surgical complications if undergoing reconstructive surgeries after cancer treatment. Among patients with breast cancer or head and neck cancers, smoking was associated with significantly higher risk of flap complications and delayed healing,[142-145] and evidence suggested that complication risk may be reduced by preoperative smoking cessation.[142,145] A systematic review including 19 studies (all retrospective except for one prospective study) of the impact of smoking on head and neck reconstructive surgery with free vascularized tissue flap showed that smoking was not significantly associated with flap failure but was associated with increased risk of hematoma and overall surgical complications.[146]

Radiation

Studies have shown that former smoking and current smoking during RT may decrease treatment response and increase complication rates, particularly in patients with head and neck cancers, but also in patients with cervical, lung, breast, or prostate cancers. Patients who receive RT and smoke may also have an increased risk of developing a second primary cancer. Smoking amplified the risk of second primary cancer associated with RT for initial breast, lung, or cervical cancers.[147-150]

In patients with head and neck cancer receiving RT, current smoking was associated with poorer rates of locoregional control.[89,151] In another cohort, patients with head and neck cancer who smoked during RT had lower rates of complete response and worse survival times than patients who never smoked or those who quit prior to treatment.[152] Smoking during RT in patients with head and neck cancer has also been shown to increase the rates of treatment-related complications. In patients with laryngopharyngeal cancers, smoking during treatment was associated with significantly elevated incidence of osteoradionecrosis and hospitalization during treatment.[153] Another study demonstrated a significantly greater decline in several HRQOL measures in patients who smoked during therapy compared with patients who quit beforehand.[154]

Among 3489 patients receiving RT as part of treatment for stage I or II cervical cancer, heavy smoking (defined as ≥1 pack/day) was the strongest independent factor for predicting long-term major bladder, rectal, or small bowel complications, with even light/moderate smoking (<1 pack/day) predisposing patients to small bowel complications, compared to patients who did not smoke.[155] In another study of 565 patients with cervical cancer who were receiving primary RT, patients who smoked during treatment had lower cure rates, higher frequency of RT side effects, and higher rates of severe, irreversible complications, compared to patients who did not smoke.[156]

Smoking during RT for NSCLC was associated with significantly decreased locoregional control.[157] Smoking may also decrease the efficacy of RT for prostate cancers and increase the prevalence of long-lasting treatment-related effects on the bowel and anal sphincter.[158-160] Concurrent smoking and RT increased the risk of cardiovascular disease in 4414 10-year survivors of breast cancer.[161]

Systemic Therapy

Data on the impact of smoking on chemotherapy are much more limited than that for surgery and RT, in part because smoking quantity during treatment is often left out of the medical record.[31] Many of the purported effects of smoking during chemotherapy are extrapolated from what is known about the impact of chemotherapy and smoking as individual health factors. Smoking has the potential to exacerbate the risk of anticancer drug-related pulmonary and cardiac toxicities such as cardiomyopathy and pulmonary fibrosis.[162] Combining neoplastic agents with RT while smoking may lead to further toxicity. Additionally, cancer drug side effects such as weight loss, cachexia, and fatigue may also be increased by smoking during treatment.[162,163]

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Many systemic anticancer agents result in some degree of immune suppression/compromise, and smoking during chemotherapy may further compromise immune function in an already vulnerable patient population.[31,162] Preclinical and clinical studies suggest that smoking and nicotine exposure can be detrimental to the function of both the adaptive[164-167] and innate immune system.[168-172] Similarly, cigarette smoking may increase the incidence of infection, particularly for smoking-related infectious diseases such as pneumonia and influenza.[173]

Preclinical studies also suggest a potential link between nicotine exposure and the development of chemoresistance,[174-178] although no clinical data are currently available to support these findings. This should not be a consideration in the decision to prescribe NRT, given the poor outcomes associated with cigarette smoking specifically.

Smoking can also impact the metabolism of certain cytotoxic chemotherapies and other systemic therapy. Smoking effects on cytochrome P450 enzymes may alter drug clearance time and plasma concentration, potentially impacting the efficacy of certain drugs for patients who smoke.[179] Health care professionals should consider whether patients are at risk for altered drug metabolism due to smoking and determine if medication or dose adjustments may be required.

Drugs with metabolisms that are known to be affected by smoking include erlotinib and irinotecan. Rapid drug clearance has been observed in patients who smoke who were receiving erlotinib therapy, such that higher doses may be required to achieve equivalent systemic exposure to standard dosing in patients who never smoke.[180,181] Similarly, smoking increases the clearance time of irinotecan, potentially lessening systemic exposure.[180,182] Given the narrow therapeutic index of systemic therapy for lung cancer, small changes in drug exposure due to smoking could affect treatment efficacy and patient outcomes.[180] Bendamustine metabolism is also likely to be impacted by smoking, resulting in decreased drug plasma concentration and increased concentration of its active metabolites.[183] However, smoking does not appear to alter the pharmacokinetic properties of taxane chemotherapeutics (eg, docetaxel, paclitaxel) despite its paradoxical protective effects on drug-induced neutropenia and leukopenia.[184]

Symptom Burden

Analysis from the US FDA Population Assessment of Tobacco and Health Study, including 1409 adults diagnosed with cancer, showed that both cigarette smoking and vaping were associated with greater cancer symptom burden (fatigue, pain, and emotional problems).[185] However, cigarette smoking in adults with cancer was associated with worse QOL, while vaping was not significantly associated. In a study of 947 patients who were undergoing chemotherapy and/or RT, smoking during treatment was linked to a higher overall burden of symptoms commonly experienced among patients with cancer. In analyses that controlled for age, gender, race, education, occupation, treatment, cancer site, and Karnofsky performance score, patients who smoke had a significantly higher symptom burden compared with patients who never smoke, both during treatment and 6 months afterward.[186] Smoking in patients with advanced lung cancer was associated with greater symptom burden on diagnosis and poorer HRQOL post-diagnosis.[187] Additional studies suggest that patients with cancer who smoke may experience more severe or frequent pain and lower quality of life than nonsmoking counterparts.[188-193]

Benefits of Smoking Cessation for Patients with a Cancer Diagnosis

For many who smoke, the benefits of smoking cessation can be appreciated immediately through reduced blood carbon monoxide levels, decreased irritative respiratory symptoms (eg, cough, shortness of breath), and improved lung function. In the long term, cessation is associated with reduced risk of smoking-related disease, development of malignancy, and smoking-related mortality.[194] Although the deleterious effects of smoking

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after a cancer diagnosis are well documented and understood, research on the benefits of cessation post-diagnosis is much more limited.[20,195] For patients with cancer, the potential benefits and risk reductions associated with cessation are of critical importance.

Studies have evaluated the impact of abstaining from smoking at or near the time of a cancer diagnosis by comparing outcomes of patients who smoke during cancer treatment to those who quit prior to treatment. Studies generally show that patients who recently quit have survival outcomes intermediate to that of patients who never smoke and those who smoke, suggesting a measurable benefit of cessation post-diagnosis. The data to support this survival pattern are derived primarily from retrospective cohorts of patients with lung or head and neck cancers,[10,11,97,111,128,152,196-201] but similar patterns have been observed for colorectal, bladder, and kidney cancers, as well as various other disease sites.[11,73,202,203] One prospective study showed that including a smoking cessation intervention within 6 months of cancer diagnosis resulted in improved survival outcomes both short- and long-term.[204]

Retrospective studies have estimated a roughly 25% reduction in risk of mortality between those who smoke and those who used to smoke who were diagnosed with NSCLC and bladder cancers.[74,197] A prospective cohort study including 212 patients with kidney cancer and who smoked showed that quitting smoking was associated with better overall mortality (hazard ratio [HR], 0.51; 95% CI, 0.31–0.85), better cancer-specific mortality (HR, 0.54; 95% CI, 0.31–0.93), and lower risk for disease progression (HR, 0.45; 95% CI, 0.29–0.71) for patients who quit smoking, compared to those who continued to smoke.[205]

A study of cancer registry data from Japan (n = 30,658) compared the risk of ACM in patients who never smoke, who formerly smoked (>3 years), who recently quit (≤3 years; 85% within the past year), and who currently smoke.[206] When combined results were weighted by disease site prevalence, risk of death in patients who recently quit was reduced by 11% compared with patients who currently smoke (HR, 0.89; 95% CI, 0.81–0.97). Risk of death in patients who never/formerly smoked was found to be 15% to 16% lower than in patients who currently smoke. Over the course of 10 years, adjusted survival rates were consistently higher among those who recently quit, formerly smoked, and never smoked. When comparing patients who recently quit smoking to those who currently smoke, a small but significant difference in survival rate was observed in those who recently quit (+1.8% to 2.9% over 5 years).[206]

A prospective longitudinal study examined the impact of smoking at diagnosis in 5185 patients with cancer across 13 disease sites over the course of at least 12 years.[11] In this study, patients who recently quit were examined as a specific subset of those who quit within 1 year of the study's structured smoking assessment, allowing for comparisons to individuals who continued to smoke during cancer treatment. For disease sites with larger recent quit cohorts (ie, lung, head/neck cancers), those who recently quit had lower overall mortality risk compared with those who continued to smoke (current smoking vs. recent quitting: lung cancer HR, 1.38–1.42; head/neck cancer HR, 2.11–2.15).[11] Similarly, two systematic reviews including studies of patients with lung cancer showed that quitting smoking was associated with a survival benefit.[10,199]

In a longitudinal prospective cohort study including 517 people who smoked with early stage (IA–IIIA) lung cancer, patients who quit smoking after diagnosis were compared to patients who continued to smoke.[13] Median OS (6.6 vs. 4.8 years, respectively; *P* = .001), 5-year OS (60.6% vs. 48.6%, respectively; *P* = .001), and 5-year progression-free survival (PFS) (54.4% vs. 43.8%, respectively; *P* = .004) were higher for patients who quit smoking, compared to patients who continued to smoke. In patients who quit smoking, a 33% decrease in overall mortality risk and a 25% decrease in cancer-specific mortality risk were observed over a mean

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7-year follow-up. The benefits of smoking cessation were significant, regardless of cancer stage, pack-years smoked, and treatment received. Analyses from an observational cohort study including 646 smokers diagnosed with NSCLC showed that quitting smoking at time of diagnosis was associated with reduced mortality at 2-year follow-up (HR, 0.75; 95% CI, 0.58–0.98), regardless of cancer stage.[207]

Abstaining from smoking at or near cancer diagnosis appears to reduce treatment-related complications compared to patients who continued smoking. In patients undergoing lung cancer resection, studies suggest that preoperative abstinence mitigates the risk of pulmonary complications and in-hospital mortality.[133,135] In one study, risk-adjusted ORs for mortality and pulmonary complications decreased as preoperative cessation time increased from 14 days to 1 month, to 1 to 12 months, and to more than 12 months.[135] A larger retrospective study comparing preoperative cessation intervals of increasing lengths suggested that longer periods of preoperative cessation may be more effective than shorter periods for reducing risk of postoperative complications, although cessation for any amount of time prior to surgery significantly reduced risk.[133] A retrospective study of 188 patients undergoing reconstructive surgery after treatment for head and neck cancer revealed that preoperative smoking cessation of at least 3 weeks led to lower incidence of wound healing complications than patients who continued smoking.[208] In a study of 107 older adult males with lung cancer who underwent elective thoracoscopic radical lung resection, smoking cessation at least 3 weeks before surgery was associated with better pain outcomes than smoking cessation <3 weeks before surgery.[209]

Abstinence has been shown to lead to improvements in various measures of general health and well-being for patients with cancer. Smoking abstinence improved performance status at 6 and 12 months post-lung cancer diagnosis over that of continued smoking when adjusting for disease stage, patient demographics, therapy, and comorbidity.[14] Additionally, patients with cancer who quit smoking benefited from lower rates of smoking-related cardiovascular and pulmonary disease.[210]

Evaluation and Assessment of Patient Smoking

These Guidelines highlight the importance of evaluating and assessing smoking status and history in patients with cancer. The AACR emphasized in a policy statement the need for universal assessment and documentation of tobacco use by patients with cancer both in the standard clinical setting and in oncology clinical trials.[20] The NCI-AACR Cancer Patient Tobacco Use Assessment Task Force published proposed core and extension items to be used for the assessment of tobacco use in patients with cancer enrolled in research trials.[211] Current practice is suboptimal, as inadequate or inconsistent assessment and documentation of smoking status has been reported both in the care setting and in the context of cancer registries and clinical trials.[23,32,212]

Despite the demonstrated adverse effects of smoking during cancer treatment, a large proportion of cancer clinical trials do not collect adequate, up-to-date information regarding patient smoking status and history, particularly for malignancies other than well-known tobacco-related cancers (eg, lung, head and neck cancers).[31,32] Such assessments are needed to make evidence-based determinations of the impact of smoking on patients, treatment efficacy, and side effects.

In a large study conducted at an NCCN member institution, a smoking assessment questionnaire was integrated into the EHRs in order to automatically identify and refer appropriate candidates for onsite cessation services.[213] The smoking assessment items incorporated into the EHR were refined based on analysis of responses from an initial patient screen containing 23 items. Response analysis revealed that the most effective questions for generating referrals included whether 1) patients smoked

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cigarettes every day, some days, or not at all; and 2) if/what other types of tobacco products were used. For patients who formerly smoked, it was important to assess the last time a patient smoked a cigarette, "even a puff," and for established enrollees to the cessation program, what type(s) of cessation aids were being used. The study revealed that just three assessment questions made it possible to efficiently and accurately identify the vast majority (>98%) of patients who currently smoke or those at risk for smoking relapse. These questions are as follows: 1) Have you smoked at least 100 cigarettes in your entire life? (yes, no); 2) Do you now smoke cigarettes every day, some days, or not at all?; and 3) Do you currently use any other tobacco products, such as cigars, pipes, chewing tobacco, snuff, dip, snus, clove cigarettes, kreteks, or bidis? (Every day/some days/not at all).

Determining Smoking Status

The NCCN Guidelines for Smoking Cessation advocate for smoking status to be updated in the patient's health record at regular intervals to indicate any status changes or quit attempts. To do so, the Panel recommends health care professionals initially ascertain: 1) whether the patient has ever smoked, and if so, then regularly assess; 2) whether the patient currently smokes; and 3) whether the patient has smoked within the past 30 days (as an arbitrary number to identify patients at very high risk of relapse). All information regarding smoking status of all tobacco products (including pipes, cigars, hookah, cigarillos, e-cigarettes/vaping, and smokeless tobacco) should be recorded in the medical record. As a follow-up to the initial evaluation, these guidelines direct health care professionals to a tailored patient assessment based on smoking status and history. Specific algorithms for patients who currently smoke (patient smoked within the last 30 days) and formerly smoked who recently quit (more than 30 days to 1 year prior) are included. For patients who never smoked or those who quit more than 1 year prior, health care professionals should urge patients to remain abstinent from smoking, explaining the benefits of remaining abstinent. For recommendations regarding lung cancer screening for current/former smoking, see the NCCN Guidelines for Lung Cancer Screening (available at www.NCCN.org).

Assessment of Current Smoking

In patients who currently smoke or use tobacco products (or those who have smoked in the last 30 days), health care professionals should assess tobacco use to understand the chances for success and risk of relapse and document the findings in the patient's health record. To assess nicotine use, health care professionals should query patients regarding the amount currently smoked or used per day (including cigarettes, pipes, cigars, hookah, cigarillos, e-cigarettes/vaping, and smokeless tobacco), the typical amount smoked or used, and how soon the patient smokes or uses tobacco products after waking up in the morning (ie, within 30 minutes). Though smoking within 5 minutes of waking is indicative of greater dependence,[214] time to first cigarette is used to determine the appropriate strength of nicotine gum and lozenge (2 or 4 mg). The Fagerstrom Test for Nicotine Dependence is an alternative standardized tool for assessing nicotine dependence.[215] However, the Panel has opted to recommend a more streamlined assessment for use in the oncology setting.

To best tailor treatment, health care professionals should also gather information regarding the patient's history of quit attempts and why they were or were not successful. Specifically, health care professionals should ascertain the longest period of abstinence achieved, the date of the most recent quit attempt, what cessation aids were used, and why these failed. It is important to document the patient's previous experience with smoking cessation aids, including any medications, behavior therapy, e-cigarettes, quitlines, websites, smart phone apps, or other media aids. The patient's reasons for why these aids were unsuccessful are important pieces of

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information. For example, the patient may have experienced side effects and/or continued cravings with medication, and this may indicate a need for alternative or complementary treatment. If the medication seemed to be ineffective, then duration of use should be noted, as it is possible that it was discontinued too soon. In this discussion, health care professionals should avoid discouragement by normalizing previous failed attempts by counseling the patient that it is the norm to undergo multiple attempts before quitting permanently and that, with each try, something new is learned.

As health care professionals advise patients to quit, the Panel recommends providing patients with reasons and ideas for smoking cessation, emphasizing the importance of both encouragement and directness with patients who smoke. When incorporating motivational interviewing (MI) to promote willingness to quit, the Panel emphasized the importance of the following general principles: 1) express empathy, 2) develop discrepancy (ie, ask patients to identify conflicts between their values/priorities and current tobacco use), 3) roll with resistance, and 4) support self-efficacy.[216,217] For a summary of the methods and data on MI for smoking cessation, see the *Principles of Behaviorial Treatment of Smoking* section below.

If patients are not ready to quit, health care professionals should assess and address patient-reported barriers and concerns regarding cessation. When possible, health care professionals should work with patients to set a near future quit date (ie, 1–3 months, or longer for patients who may take longer to move from precontemplation to contemplating quitting[218]) and/or consider smoking reduction with pharmacotherapy and counseling with the goal of achieving and maintaining abstinence from smoking in the future.[219] Health care professionals should discuss the impact of continued smoking on cancer treatment outcomes and complications from chemotherapy, radiation, and surgery. Starting pharmacotherapy prior to having a quit date is also appropriate in these patients.[220] A meta-analysis of 10 randomized trials in 3760 patients with cancer found quit rates to be comparable when comparing abrupt cessation to gradual smoking reduction.[221] Therefore, both options can be used after discussions with the patient. Trial data since then have continued to mirror this trend of comparable success rates with abrupt cessation and gradual reduction.[222] At each visit, health care professionals should reassess readiness to quit and engage in motivational dialogue as indicated. Even brief advice (ie, about 3 minutes) has been shown to be beneficial.[62,63,223,224]

Assessment of Former Smoking

Health care professionals should evaluate and document the risk of relapse for patients with more than 30 days' abstinence. The Panel suggests the following characteristics to identify patients at high risk for relapse: frequent/intense cravings; elevated anxiety, stress, or depression; current or prior history of psychiatric disorders[225]; cohabitating or working with someone who uses tobacco; quitting within the past year; recently initiated smoking cessation pharmacotherapy or other nicotine delivery mechanisms; drug/alcohol use or misuse[226]; recent history of nicotine use disorder; and/or low self-confidence in ability to quit/maintain abstinence. Patients with poor pain control may also be at increased risk for smoking relapse.[227] There is also an increased risk of relapse among patients living with HIV.[228] The Panel considers patients demonstrating at least one of these characteristics to be at higher risk for relapse and recommends a management plan tailored to prevent relapse. Health care professionals should discuss risk of relapse with patients and provide guidance and support to promote continued smoking cessation attempts. As indicated, health care professionals, in addition to their efforts to assess smoking treatment, may refer patients with psychiatric and substance use disorders to a specialist (see the NCCN Guidelines for Distress Management, available at www.NCCN.org).

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Treatment of patients who demonstrate an elevated risk of relapse includes behavior therapy with counseling on relapse risk factors (eg, identification of triggers for smoking urges or living with someone who smokes) and relapse prevention. Pharmacotherapy can be considered to promote maintenance of abstinence. Health care professionals should review smoking-associated risks for patients with cancer as well as the health benefits of abstinence. All management plans and counseling should be documented in the patient health record. For patients deemed to be at low risk for relapse, health care professionals should reinforce success and highlight the importance of continued abstinence.

It is important to regularly reevaluate patients' smoking status and risk of relapse. If relapse occurs, patients should be evaluated per the recommendations for current smoking. Additionally, health care professionals should remain aware that patient self-report of smoking status might underestimate the rate of current smoking among patients with cancer, as is evidenced by research comparing self-reported and objective measures.[229-231] Patients who remain abstinent should regularly undergo reevaluation (ie, at each visit) with documentation of any risk factor changes.

Treatment of Smoking During or After Cancer Treatment

The following recommendations are appropriate for patients who are currently undergoing cancer treatment as well as cancer survivors.

Devising a Treatment Plan

Following assessments, health care professionals should establish a personalized quit plan for each patient that takes into account the patient's nicotine dependency, prior quit attempts and any cessation aids used, and smoking treatment options. Health care professionals should work with patients to set a quit date as soon as possible. Risks of relapse and slips should be discussed with the patient along with guidance and support for continued abstinence from smoking.

Smoking Cessation and Cancer Surgery

Smoking has been shown to increase the risk of pulmonary and cardiovascular complications, surgical site infection, and poor wound healing in patients undergoing surgery. For an overview of the data, see the section on *Smoking-Related Effects on Treatment Efficacy, Side Effects, and Outcomes* in this Discussion. Preoperative pharmacotherapy options for patients within 1 week of surgery include combination NRT (preferred; category 1) or varenicline. NRT is listed as preferred in this setting because guidelines for its use include a quit date within 24 hours, while varenicline quit dates are typically 7 days from initiation of therapy. However, in some clinical situations, varenicline has been used with an earlier quit date. If there is 1 week or more until scheduled surgery, combination NRT or varenicline are both category 1 recommendations.

NRT is not a contraindication to surgery. Some surgeons may not allow perioperative NRT use in their patients. This practice may be due to findings in animal studies regarding an association between nicotine and decreased wound healing. However, no similar data in humans exist, and NRT typically provides a lower overall dose of nicotine compared with cigarettes, while nearly doubling the chance of achieving smoking abstinence.[232] NRT at the normal doses has not been shown to negatively affect acute wound healing,[233,234] and offers benefits over continued smoking.

Preoperative smoking cessation reduces postoperative complications. In randomized trials of smoking cessation or reduction interventions, postoperative complications from orthopedic surgery were significantly decreased in patients who quit compared with patients who continued smoking.[235,236] Study findings also support the benefits of preoperative smoking cessation in patients with cancer, which has been shown to

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reduce postoperative morbidity for various cancer types.[132,133,135,237,238] The optimal timeframe for abstinence from smoking to achieve full benefit is not yet clear, but longer periods of time are likely to be more effective for reducing postoperative complications.[132,133]

Patients who smoke who need surgery should be treated with evidence-based approaches, including pharmacotherapy and behavior therapy, in order to improve surgical outcomes and long-term cessation rates.[235,236,239,240] For patients with planned cancer surgery, abstinence from smoking should occur as soon as possible before surgery, regardless of how soon surgery will be occurring. There is no evidence to support a harmful effect of a quit attempt at any point in time before surgery. Even quitting on the day of surgery has a benefit. Although longer periods of preoperative abstinence from smoking may confer better surgical outcomes,[132,133,238,241] the Panel emphasizes that patient smoking should not delay appropriate timing for cancer resection, and access to cancer surgery should not be restricted solely on the basis of continued smoking.[242]

Elective procedures, such as plastic surgery reconstruction, may be associated with benefit from delaying surgery for a period of time after smoking cessation. Consensus on an optimal time period of preoperative cessation has not yet been demonstrated through the existing literature.[243] Providing surgery-specific resources and advice for smoking cessation may facilitate smoking reduction or cessation in patients undergoing elective (non-cancer) surgery.[244,245]

Primary Treatment

Based on clinical trial data of smoking cessation in patients with cancer, the Panel recommends a combination frontline approach including pharmacotherapy and behavior therapy for smoking cessation for patients with cancer. Population studies and meta-analyses of randomized or quasi-randomized trial data support the addition of behavior therapy to pharmacotherapy to enhance the rate of success.[246-249]

Preferred primary therapy options in these Guidelines are combination NRT (category 1) that uses long-acting NRT (nicotine patch) plus a short-acting NRT, such as nicotine gum, lozenge, inhaler, or nasal spray. An alternate option is oral varenicline (category 1). Patients commonly underdose when using combination NRT.[250-252] Patients should be advised that, as with any medication, adverse events may occur with nicotine replacement pharmacotherapy, but these are typically mild and short-lived and can be managed by changing the dose or product. For discussion of the evidence for and safety of individual pharmacotherapeutic regimens, see the section below on *Principles of Pharmacotherapy*.

Follow-up

Assessment of effectiveness of counseling and pharmacotherapy, smoking status, and adverse effects of pharmacotherapy should be performed in person or by phone/telehealth at a minimum of: within 3 weeks of initiating treatment for smoking (within 1 week preferred, if possible), and following 12 weeks of therapy. Assessment should continue on a periodic basis moving forward, including within about 12 weeks following completion of pharmacotherapy. Most nicotine withdrawal symptoms manifest in the first 1 to 2 days, peak within the first week, and subside within 2 to 4 weeks of quitting. Increased appetite, weight gain, and cravings may persist for several months.

Slips commonly happen in the first week of abstinence. Patients who do not quit immediately may quit at a later point. Therefore, health care professionals should encourage continued treatment adherence through and beyond brief slips, with adjustments to dose or behavior therapy frequency as indicated.[253] Adverse effects may also warrant dose adjustments. It is important to parse out perceived withdrawal symptoms

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from treatment side effects. It also is important to assess for drug interactions, as the drug levels of certain medications may be altered when a person stops smoking. A comprehensive list of drug interactions with tobacco smoke has been published: https://smokingcessationleadership.ucsf.edu/sites/smokingcessationleadership.ucsf.edu/files/A4%20DI%20TABLE.pdf

When possible, in-person follow-up during planned clinical visits or individual/group therapy sessions is preferred. To minimize the burden on patients in active cancer therapy, behavior therapy can be provided by a trained member of the health care team during oncology visits. Alternatives include phone or telehealth. During follow-up, health care professionals should assess risk of relapse and as indicated, consider adjusting the dose and or type of pharmacotherapy. Patients may slip or relapse, which is expected and can be managed. Maintain close follow-up through the duration of therapy. At 12 weeks, assessment of smoking status should be made in person or by phone/telehealth. For pharmacotherapy courses exceeding 12 weeks duration, assessment should be repeated at the end of the course of therapy.

For patients who remain abstinent, additional follow-up should take place at 6 and 12 months, either in person or by phone/telehealth. Motivational strategies should be used to promote continued abstinence. Duration of pharmacotherapy can be extended beyond 6 months if clinically indicated to maintain abstinence. For patients who experience smoking relapse, treatment with either combination NRT or varenicline should be re-initiated.

Treatment for Relapse or Sustained Smoking

For patients who continue to smoke or experience relapse, assess effectiveness or prior pharmacotherapies and medical adherence (ie, frequency and proper usage). The Panel recommends one of two options in this setting. The first option is to continue or resume primary therapy, reinforce proper use of medication, and provide additional behavior therapy. The second option is switching to the alternate preferred option (combination NRT or varenicline). The Panel recommends that these regimens be paired with additional behavior therapy. In most circumstances, both preferred primary therapy approaches (combination NRT and varenicline) should be tried before proceeding to any other pharmacotherapy options. However, bupropion (long-acting) with or without NRT may be appropriate in earlier treatment settings for select patients (eg, those with symptoms of depression or fatigue). For those being treated with varenicline, the addition of NRT or bupropion (long-acting) can be considered as clinically indicated to maintain abstinence despite continued urges to smoke. The decision to continue or switch therapy should be based on prior cessation success, patient preference, toxicity, cost and/or coverage, and/or a change in clinical status (eg, upcoming surgery).

After switching therapy or adjusting the dose of an existing regimen, follow-up should occur within 3 weeks (within 1 week is preferred) and after 12 weeks of therapy. Smoking status should be re-evaluated. For patients who remain abstinent, the course of pharmacotherapy can be extended as clinically indicated. Additional follow-up at 6 and 12 months after successful quitting is recommended. For further relapse or sustained smoking, extended duration of pharmacotherapy or switching to an alternate regimen can be considered. Additional or more intensive behavior therapy is also an option.

Principles of Pharmacotherapy

General

A minimum of 12 weeks of pharmacotherapy is recommended for the initial quit attempt. Evidence supports extending the duration of therapy to 6 months.[254-256] Therefore, 6 months of treatment is at least minimally

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preferred. However, duration of therapy may even be extended to ≥1 y to promote continued abstinence, for example while patients are undergoing active cancer therapy versus those who are long-term survivors.[257]

Follow-up is recommended within 3 weeks of starting pharmacotherapy to coordinate with scheduled oncology appointments, though follow-up within 1 week is preferred. For relapse or sustained smoking, options include continuation of the initial agent or a switch to the alternative preferred agent. Health care professionals should educate patients to take smoking treatment medications according to the prescribed regimen and for the full course of therapy. Dose adjustments should be considered as clinically indicated. Attempts at smoking reduction should be tracked. If reduction efforts stall or if complete abstinence seems unlikely, health care professionals should consider an alternative pharmacotherapy regimen.

In most circumstances, the side effects related to primary smoking treatment medications, which often peak in the first week of administration, are minimal and are considered an acceptable risk compared to smoking. A review of post-marketing case reports on adverse neuropsychiatric effects from smoking treatment medications has generated some safety concerns in the past,[258] but large-scale analyses of the data support the safety of these regimens.[259,260] Although serious side effects of primary treatment approaches are extremely rare, health care professionals should refer to manufacturer inserts for exhaustive lists of potential side effects and warnings.

Adherence to pharmacotherapy is important to promote optimal outcomes and success, and numerous studies have tested interventions designed to promote and improve medication adherence.[253,261] Nonadherence is associated with perceiving that cessation aids are not needed, adverse effects, forgetfulness, and limited finances.[262,263]

Below, data from various clinical trials are discussed. Included in this Discussion are findings from a Cochrane network meta-analysis that included data on pharmacologic interventions across 267 individual studies in 101,804 participants.[264] The authors characterized positive treatment outcome as continuous or prolonged abstinence at least 6 months from the start of smoking treatment. Harm outcomes were measured by the incidence of serious adverse events associated with treatment.

Pharmacotherapy Options

For patients with cancer, the Guidelines recommend primary therapy with either combination NRT (long-acting patch plus a short-acting formulation) or varenicline, paired with behavior therapy. If smoking is sustained or relapse occurs while a patient is on an initial primary therapy regimen, health care professionals should continue therapy with that initial regimen or switch to the alternate primary therapy option. Both preferred primary therapy options (combination NRT or varenicline) should be used before trying other pharmacotherapy options. Bupropion with or without NRT may be appropriate for select patients (eg, those with symptoms of depression or fatigue). For those being treated with varenicline, the addition of NRT or bupropion can be considered as clinically indicated to maintain abstinence despite continued urges to smoke. Data supporting the recommended pharmacotherapy options are described below.

Varenicline

Efficacy

Varenicline is a non-nicotinic partial agonist of the alpha4beta2 subtype of the nicotinic acetylcholine receptor. Varenicline partially mimics the effects of nicotine in the brain's reward center and competitively inhibits the binding of nicotine from cigarettes.[265]

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Systematic reviews/meta-analyses have identified varenicline as the most effective single pharmacotherapy option for smoking cessation.[264,266,267] However, medication adherence is an important factor in cessation success.[253] Cochrane network meta-analysis data report that varenicline increases the odds of smoking cessation by almost three-fold compared with placebo (OR, 2.88; 95% CI, 2.40–3.47).[264] Direct comparison of the cumulative data suggest that varenicline was more efficacious than bupropion (OR, 1.59; 95% CI, 1.29–1.96) and single forms of NRT such as nicotine patch, nicotine gum, and other formulations (OR, 1.57; 95% CI, 1.29–1.91).[264] Varenicline appeared to be equally as likely to promote smoking cessation as combined treatment with more than one form of NRT (OR, 1.06; 95% CI, 0.75–1.48), so that both may be offered depending on patient circumstances.[264] An analysis from a Cochrane review including 11 studies showed that varenicline prevents relapse in those who abstain from smoking, based on moderate-certainty evidence (RR, 1.23; 95% CI, 1.08–1.41).[268]

Results published from the double-blind EAGLES RCT (n = 8144) revealed that patients treated with varenicline achieved higher abstinence than patients receiving placebo (OR, 3.61; 95% CI, 3.07–4.24), nicotine patch (OR, 1.68; 95% CI, 1.46–1.93), or bupropion (OR, 1.75; 95% CI, 1.52–2.01).[269] A study that investigated the efficacy of varenicline specifically for patients with cancer revealed 84% retention and 40% abstinence at 12 weeks. Side effect profiles mimicked those observed in the general population, and abstinence improved cognitive function and reduced negative effect over time.[270]

Varenicline may also be efficacious for smoking reduction in those not ready to quit. A clinical trial enrolling 1510 individuals revealed that a 24-week course of varenicline effectively promoted smoking cessation in patients who were unwilling to quit but willing to gradually reduce cigarette consumption.[271] This study provided evidence that this agent may be an alternative for patients who cannot or will not attempt abrupt cessation. A clinical trial in 1236 individuals who smoke showed that an additional 12 weeks of varenicline maintenance therapy helped to sustain continued abstinence in those who successfully quit during initial treatment.[256] A 2021 clinical trial among 1251 treatment-seeking individuals found that extended therapy of varenicline for 24 (vs. 12) weeks did not produce higher abstinence rates. Taken together, results of these trials suggest that tailoring extended varenicline therapy to individuals at high risk of relapse may be the most appropriate approach.[272] Additionally, another RCT showed that varenicline was effective and well tolerated for retreating patients who had previously received this agent (n = 498).[273]

Several studies have also suggested that varenicline dose increases beyond the standard 2 mg/day may boost treatment efficacy in patients who had a low or no response to standard dosing.[274-276] A double-blind RCT of 503 patients who smoke found no evidence to suggest that gradual dose titration beyond the standard 2-mg dose (up to a maximum 5 mg/day) lessened frequency of urges and nicotine withdrawal symptoms, or increased cessation rates.[274] However, dose increases exacerbated adverse effects such as nausea and vomiting for some individuals.[274,276] A propensity score-matched analysis including 214 patients who reduced smoking by 50% but were unable to quit showed that patients who received an increased dosage of varenicline (3 mg/day; n = 72) reported greater 7-day abstinence, compared to patients who maintained the standard dose (2 mg/day; n = 142) at 3- (RR, 2.9; 95% CI, 1.1–4.7; *P* = .004), 6- (RR, 2.5; 95% CI, 1.1–5.2; *P* = .02), and 9-month follow-up (RR, 2.7; 95% CI, 1.2–5.2; *P* = .008), based on models adjusted for imputation and propensity score matching.[275] In patients who have reduced smoking but not yet quit, a dosage increase to 3 mg/day may be considered, if tolerated.

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Safety

Varenicline safety has been extensively examined to determine the risk of adverse effects, particularly serious cardiovascular events and neuropsychiatric changes. Initial phase III studies found varenicline to be safe and generally well tolerated compared with bupropion or placebo; common side effects included nausea, insomnia, and abnormal dreams with rates of approximately 28% to 29%, 14%, and 10% to 13%, respectively.[277,278]

Concerns regarding neuropsychiatric adverse effects of varenicline have been extensively investigated in patients who smoke and have comorbid psychiatric disorders.[264] Despite reviews of case reports that raised concern,[258] a 2015 systematic review and meta-analysis of 39 randomized controlled smoking cessation trials identified no evidence to suggest that varenicline increases risk of suicide or suicide attempts, suicidal ideation, depression, or death.[260] Another trial showed that varenicline increased smoking cessation rates without exacerbating anxiety and depression symptoms in adults with stably treated current or past depression.[279] Results were published from a large double-blind RCT (EAGLES trial) that enrolled two cohorts: individuals with psychiatric disorders (n = 4116) and those without psychiatric disorders (n = 4028). No significant increase in neuropsychiatric events was observed for varenicline relative to nicotine patch or placebo. Varenicline was associated with significantly higher abstinence than bupropion plus nicotine patch, as well as placebo.[269]

Cardiovascular risks have also been examined. Importantly, systematic reviews and meta-analyses of RCT data have not identified a significant link between varenicline and increased risk of serious cardiovascular adverse events.[264,280-282] However, the cardiovascular safety of varenicline has remained a topic of interest and concern,[283-285] although the cardiovascular risks of continued smoking has been extensively documented.[70]

In a 2015 retrospective review of 164,766 individuals who received pharmacotherapy for smoking cessation (varenicline, n = 51,450; NRT, n = 106,759; bupropion, n = 6557), neither varenicline nor bupropion posed an elevated risk of cardiovascular or neuropsychiatric (ie, depression, self-harm) events compared with NRT.[286] Based on the current evidence base for safety risks, the Panel considers varenicline to be safe and to have a favorable risk/benefit ratio for use in patients with cancer who smoke.

Although rare, elevated seizure risk can be a concern in certain individuals receiving varenicline therapy.[287,288] In patients with a history of seizure disorder or at risk for seizures (eg, patients with brain metastases or primary brain tumors), varenicline should be used with caution. Dose reduction is indicated if there is renal impairment.

Combination Nicotine Replacement Therapy

Efficacy

NRT offers an alternative nicotine delivery method and can be used to ameliorate nicotine withdrawal symptoms during cessation attempts. Combination NRT incorporating long-term and short-acting NRT offers the greatest potential benefits for those who smoke.[239,264,268,289,290] Multiple Cochrane reviews show that, compared with single forms of NRT, combination NRT using a patch plus short-acting NRT improved the odds of quitting.[246,264,291,292] Data show that all forms of NRT are superior to placebo,[293] but people who smoke using combination NRT were almost three times as likely to succeed (OR, 2.73; 95% CI, 2.07–3.65).[264] Regarding duration of use, a 2015 study of 525 individuals seeking treatment for smoking found that NRT duration of 24 weeks was more efficacious than 8 weeks; however, NRT use beyond 24 weeks was safe but did not result in greater efficacy.[257] This study focused on NRT monotherapy and did not vary in duration of dual-use preparations.

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The success of NRT is bolstered by concurrent behavior therapy to support cessation. In a large population study, over-the-counter NRT resulted in similar rates of cessation to those who used no aid.[294] The addition of behavior therapy to NRT increased the odds of success nearly threefold.

Safety

The safety of combination NRT for use in humans, including long-term use, has been demonstrated and benefits are considered to outweigh potential risks.[295] Importantly, health care professionals should be aware that blood nicotine levels from NRT, including combination NRT, are significantly less than that from smoking cigarettes.[296-298] In fact, patients commonly underdose when using combination NRT,[250-252] and although significant nicotine toxicity is possible, it is rare and usually short-lived. Therefore, health care professionals and patients who smoke should not be dissuaded from using NRT to foster quitting and long-term abstinence. Reviews of the data suggest that NRT is not linked to increased serious cardiovascular adverse events when used for smoking cessation.[282,299] While myocardial infarction has rarely been reported in patients who use NRT, there is insufficient evidence that NRT increases the risk of myocardial infarction or cardiovascular disease.[291,293,299,300] Data from large case series have not shown elevated risk with the use of NRT in patients with acute coronary syndromes.[301,302]

In the past, the safety of NRT has been evaluated in light of the bioactivity of nicotine and evidence that this drug can promote cell growth in certain types of cancer cells.[303] Some in vitro data suggested that nicotine increased the malignant potential of small cell lung cancer cells[304]; induced chemoresistance in models using lung cancer cells[174-176] and nasal epithelial cells[177]; and promoted chemoresistance and metastasis in pancreatic cancer cell and mouse models.[178] However, other studies suggested no effects of physiological levels of nicotine exposure on tumorigenesis in mouse lung cancer models.[305,306] Moreover, there is no evidence from human studies that NRT causes cancer in humans.[305-309] Evaluation of data from 3320 participants in the Lung Health Study, which recorded in-study NRT use and smoking exposure, found that NRT was not a significant predictor of lung cancer, while smoking was.[308]

Varenicline + NRT

A study in 435 individuals who smoke found that the addition of nicotine patch to varenicline therapy significantly increased the cessation rates at the end of treatment (12 weeks), at 24 weeks, and at 6-month follow-up.[310] No significant differences were noted for side effect incidence between varenicline/NRT and varenicline/placebo with the exception of skin reactions, which were increased with combination therapy (14.4% vs. 7.8%; $P = .03$). However, one RCT of 341 individuals who smoke did not find enhanced cessation rates at 12- and 24-week follow-up among individuals receiving a combination of varenicline and nicotine patch, versus varenicline alone.[311] The addition of nicotine patch to varenicline did not cause significant changes in side effect profiles. Similarly, a trial in 117 participants did not find evidence that the addition of nicotine patch to varenicline increased abstinence at 1, 4, or 12 weeks after the targeted quit date, and no between-group differences in adverse effects were found.[312] Based on the evidence, varenicline combined with NRT may be considered as a pharmacotherapy option. Though the studies described above specifically examined the nicotine patch,[310,311] any form of NRT or combination NRT (ie, nicotine patch + short-acting NRT) may be used.

Bupropion + NRT

A large trial in the United Kingdom (n = 1071) examined the efficacy of NRT alone, bupropion alone, and NRT plus bupropion.[313] All participants received 7 weeks of behavior therapy support in addition to the pharmacologic interventions. Abstinence at 6-month follow-up ranged from 24.2% to 27.9% and did not differ significantly between cohorts. Several

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unwanted side effects were more common with bupropion than NRT (eg, disturbed sleep, dry mouth, headaches, nausea), and side effects of combination therapy were not significantly different versus bupropion alone. Five serious adverse events occurred in the bupropion group, including allergic reaction (n = 3), neuropsychiatric symptoms (n = 1), and chest pain (n = 1). A trend toward improved efficacy of bupropion in patients with a history of depression was noted ($\chi^2 = 2.86$; $P = .091$).

A double-blind RCT compared bupropion + NRT, bupropion alone, nicotine patch alone, and placebo in 893 individuals who smoked at least 15 cigarettes per day. At 12 months, the highest abstinence rates were observed for the bupropion + NRT group (35.5%) and bupropion only group (30.3%), although these groups did not differ significantly.[314] A smaller RCT studying the addition of bupropion to combination NRT and behavior therapy in patients with schizophrenia suggested that combination pharmacotherapy promoted smoking reduction and cessation, but also demonstrated a high relapse rate after discontinuation of treatment.[315] A 2014 meta-analysis of 12 trials examining this combination revealed a nonsignificant trend in improved cessation with the addition of NRT to bupropion.[316]

Bupropion

Bupropion was first approved to treat depression but its efficacy as a cessation aid also became apparent. In addition to its effects on the dopaminergic and adrenergic systems, this agent also acts as an inhibitor of nicotinic acetylcholine receptors. A 2014 Cochrane review of 44 trials examined bupropion efficacy, revealing an RR of 1.62 (95% CI, 1.49–1.76).[316] Results from the EAGLES trial (n = 8144) revealed that patients receiving bupropion achieved superior abstinence compared with placebo (OR, 2.07; 95% CI, 1.75–2.45).[269] Efficacy was similar to nicotine patch but less than that for varenicline. Some evidence suggests that bupropion may be particularly beneficial as a smoking cessation agent for persons with depression.[313,317] Additionally, longer duration of bupropion treatment may help to prevent relapse in those who have successfully quit.[318]

Bupropion is a CYP2B6 inhibitor and may increase the concentration of drugs that metabolize or that also inhibit CYP2B6 (ie, certain antidepressants, antipsychotics, beta-blockers, and Type 1C antiarrhythmics).[319] Dose reduction of these agents may be considered when used concurrently with bupropion. The selective estrogen receptor modulator tamoxifen requires metabolic activation by CYP2D6 and may be less effective when used concurrently with a CYP2B6 inhibitor such as bupropion.[320] Bupropion reduces the seizure threshold, and meta-analyses of trial data have found a 0.1% seizure risk among those receiving the drug for smoking cessation.[316] In patients with seizure risk (eg, patients with brain metastases or primary brain tumors who have a history or elevated risk of seizure), bupropion should be avoided. Bupropion is contraindicated for those taking monoamine oxidase inhibitors (MAOIs), those with a current diagnosis or history of anorexia or bulimia, or in cases of abrupt discontinuation of alcohol, benzodiazepines, barbiturates, or antiepileptic drugs. Bupropion is a norepinephrine-dopamine reuptake inhibitor; central nervous system (CNS) toxicity may occur when used concurrently with other dopaminergic drugs.

Neuropsychiatric effects have also been identified as a safety concern with bupropion, although to a lesser extent than with varenicline.[258] However, systematic reviews of the data have found that serious neuropsychiatric adverse events were rarely associated with bupropion prescribed for smoking cessation, including studies of bupropion in patients with mental illness.[264,321] In the EAGLES trial, no significant increase in neuropsychiatric events was observed for bupropion relative to nicotine patch or placebo.[269]

Regarding risk of serious adverse cardiovascular effects, meta-analyses do not show elevated risk as a result of bupropion use for smoking

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cessation.[264,282,316] Dose adjustments/reductions are indicated in the event of renal or hepatic impairment.[319]

Varenicline + Bupropion

In an RCT of individuals who smoke who demonstrated an inadequate response to front-line nicotine patch treatment (n = 222), combination therapy with varenicline and bupropion appeared to be more efficacious than varenicline alone as a second-line therapy option.[322] This observation was more pronounced among males and those with a high level of nicotine dependency. Although no significant differences in side effects were observed between varenicline and bupropion versus varenicline alone, dose reductions were required for 11.5% and 24.8% of patients, respectively. Common side effects were vivid dreams, change in taste perception, thirst, insomnia, and irritability. In a follow-up to this study, males who smoke (*N* = 174) were randomized to receive combination varenicline + bupropion or varenicline alone.[323] Like the previous study, the combination therapy was more efficacious than varenicline alone in those with a higher baseline level of nicotine dependence (n = 63; OR, 3.14; 95% CI, 1.11–8.92; *P* [one-tailed] = .016). Another study of varenicline + bupropion therapy versus varenicline alone (n = 316) demonstrated that combination therapy increased prolonged abstinence but did not affect 7-day point prevalence at 12- and 26-week follow-up, and no significant differences were observed between the groups at 52 weeks.[324] In this study, anxiety (7.2% vs. 3.1%; *P* = .04) and depressive symptoms (3.6% vs. 0.8%; *P* = .03) occurred more frequently in patients receiving combination therapy versus varenicline alone. In an RCT (n = 385), the addition of bupropion to varenicline did not improve prolonged abstinence at 12 months compared with varenicline alone, although both regimens were superior to placebo.[325] The diversity in populations used for these studies makes it difficult to draw conclusions about the efficacy of combination varenicline and bupropion, and this regimen is not commonly used as a first-line approach.

Principles of Behavioral Treatment of Smoking

General Principles

The Guidelines provide the following guiding principles on behavior therapy, which have been developed in consideration of the existing evidence base, clinical practice guidelines, and expert consensus.[246,247,295]

The Panel recommends a combination of behavior therapy with pharmacotherapy for best outcomes. In fact, studies suggest that counseling for smoking cessation may enhance patient satisfaction.[24,25] A 2012 systematic review of 41 studies provided support for the efficacy of this approach.[246] The “real world effectiveness” of adding a behavior therapy component to smoking cessation therapy was further supported by a large population study published in 2014.[294] Additionally, a 2016 meta-analysis of data from 1239 patients with head and neck cancer showed improved smoking cessation rates with the addition of counseling to usual care (NRT).[326] A systematic review published in 2023, which included 17 RCTs and 19 non-RCT studies, also supported a combined behavioral and pharmacologic approach.[248]

Behavior therapy may enhance motivation and support optimal medication strategies and adherence to pharmacotherapy. When possible, therapy should be provided by a tobacco treatment specialist or dedicated staff member (ie, nurse, medical assistant, health educator) with training in evidence-based treatment of smoking.[327]

As a general principle, the Panel recommends more intensive and sustained behavior therapy over brief advice or counseling.[247,328,329] However, brief counseling is better than no counseling. The evidence supports a measurable dose-response effect of behavior therapy with more numerous and/or longer sessions delivering improved cessation rates.[247,330,331] An RCT published in 2020 including 303 patients with cancer who also smoke cigarettes showed that those randomized to

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receive intensive treatment (free FDA-approved cessation medication and 4 weekly telephone counseling sessions, followed by 4 biweekly telephone sessions and 3 monthly booster sessions) were more likely to have quit smoking at 6-month follow-up (biochemically confirmed) than patients who received standard treatment (medication advice and 4 weekly telephone counseling sessions) (35% vs. 17%, respectively; OR, 2.15; 95% CI, 1.14–4.05; *P* = .02).[329] Among patients in the intensive treatment group, completion of more counseling sessions, but not medication use, was significantly associated with likelihood of quitting.

The Panel recommends that the first counseling session occur within 3 weeks of cessation, but preferably within 1 week as a majority of people who smoke relapse during this period.[332] As described above, longer, more frequent sessions are associated with higher abstinence rates.[328] If intensive therapy is not feasible, brief counseling of about 3 minutes should still be given. Studies have demonstrated a small but significant increase in smoking abstinence rates with brief counseling lasting only a few minutes.[62,63,223,224] As patients progress through multiple lines of pharmacotherapy, behavior therapy should be progressively intensified with referral to specialty care (eg, tobacco treatment specialist, psychologist) as indicated. Studies have also demonstrated additional benefit for relapse prevention of extending behavior therapy for ≥6 months.[333-335]

The most successful behavior therapy strategies use practical counseling in which patients learn coping skills and receive support and education. Optimally, behavior therapy plans should take into account the presence of a nicotine use disorder, previous quit attempts, and cessation aids utilized. In doing so, patients can be equipped with tailored strategies to cope with nicotine withdrawal symptoms, environmental smoking triggers (eg, coffee, alcohol, social situations), and stressful situations. For instance, the addition of a cognitive behavior therapy program designed to improve stress management improved cessation rates over controls receiving standard smoking cessation therapy.[336] The presence of other household members who smoke, or lack of household smoking rules, is a predictor of relapse or sustained smoking,[329,337,338] and there is no risk-free level of secondhand smoke exposure among those who don't smoke by causing lung cancer, heart disease, and acute respiratory effects.[7] Therefore, to benefit the patient, individuals in the household should be encouraged to abstain from smoking. Health care professionals should prepare patients for nicotine withdrawal symptoms and cravings, which manifest in the first 1 to 2 days, peak within the first week, and subside within 2 to 4 weeks of quitting.[339-341]

A number of modalities can be used to deliver behavior therapy to patients. Counseling can take place in a variety of settings such as in person, remotely by phone or telehealth,[66,342,343] or through web-based interventions.[344,345] Effective in-person counseling can occur as an individual session or in the group therapy setting.[346,347] Additionally, print materials[348] and mobile telephone "apps"[349-351] can be used to deliver behavior therapy. However, health care professionals should be aware that media-based behavioral interventions, particularly those using mobile telephones, might vary in the degree to which they comply with clinical practice guidelines.[352] Digital resources such as the NCI's smokefree.gov provide free, evidence-based cessation support through mobile-optimized websites, text messaging programs, and two mobile applications. A study revealed that approximately 7 to 8 million people accessed smokefree.gov resources in 2018 alone and that people do take advantage of the full range of technology tools.[353] Ultimately, patients should be referred to evidence-based resources for quitting.

A study investigating preferences for the provision of smoking cessation information among Canadian patients with cancer showed that patients most often preferred print materials (45%), followed by telephone support

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(39%), speaking with a clinician (29%), website-based information (15%), and support groups (11%). Patients ≤45 years were more likely to prefer cessation advice via telephone, while patients ≥46 years preferred print materials.[354] Selection of a particular modality or modalities should be guided by patient preference, medical history, and resource availability. A meta-analysis including 21 RCTs targeting smoking cessation in cancer survivors showed larger effects with interventions delivered by nurses, compared to interventions delivered by others (eg, research team, doctor, educator, counselor/therapist).[355]

For patients who are unable to quit, referral to a smoking cessation specialist is encouraged when available. If specialized resources are limited, effective behavior counseling can still be provided. For instance, brief counseling by health care professionals has been shown to generate a small but important increase in quit rates.[62-64] Additionally, quitlines can provide essential behavioral support in the absence of in-person counseling resources.[66] For instance, the addition of combination NRT to quitline counseling improved cessation outcomes.[356]

Tailoring Behavior Therapy for Patients with Cancer

As resources allow, specialized treatment centers should provide tailored treatment for smoking that address the unique needs of patients with cancer, or refer to external resources (eg, quitlines) that provide such specialized services. For patients in active cancer treatment, behavior therapy can be provided during scheduled oncology visits to obviate the need for additional appointments. Interventions initiated during hospital stays have also been successful. Intensive behavioral interventions provided during hospitalization, with at least 1 month of subsequent follow-up upon discharge, increased smoking cessation rates.[331] Beneficial services might include individual and group therapy focusing on the challenges specific to cancer treatment and survival, which would ideally be provided by clinicians experienced in working with patients with cancer. One study suggested age-based differences in preferences for smoking cessation resources.[354]

The prevalence of psychiatric disorders or serious emotional concerns in patients with cancer is high, with several large studies reporting rates between 30% and 40%.[357-359] The high rates of anxiety, depression, and stress can present a challenge for patients with cancer who attempt to quit smoking in the face of these common smoking/relapse triggers. However, data show that among those who quit, anxiety and depression improves.[360,361] Therefore, anxiety and depression should not be a barrier to aggressive smoking treatment.

Health care professionals should routinely assess for symptoms of depression, anxiety, and stress among those who smoke throughout treatment for smoking. Patients with cancer, particularly those experiencing psychiatric comorbidity, may benefit significantly from behavior therapy programs tailored to manage cancer-related issues that predispose patients to relapse. Therefore, even if patients are hesitant, smoking treatment should be encouraged. Referral for appropriate evaluation and treatment as needed or to specialized smoking treatment programs may be necessary so that these patients have access to staff trained to treat psychiatric and/or substance use disorders.

Motivational Enhancement

All patients with and survivors of cancer should be encouraged to quit. Strategies to enhance motivation to quit are beneficial for all patients throughout treatment, including those who are currently ambivalent or unwilling to quit. MI is one evidence-based approach to foster motivation to quit in which a clinician offers empathy as the patient explores ambivalence regarding quitting smoking. Clinicians reflect, validate, and summarize patient emotions and concerns and help patients identify discrepancies between smoking and core values/goals (eg, health,

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parenting) and support patient confidence to achieve and maintain abstinence.[39,216,217]

A 2015 Cochrane database review of 28 studies examined the efficacy of MI for smoking cessation, revealing a modest but significant increase in chance of quitting with MI versus brief advice or usual care.[362] MI by a primary care physician appeared to be somewhat more successful than that administered by counselors, although both were effective. Notably, one-time short MI sessions of <20 minutes had demonstrated efficacy.[362] One systematic review summarized the evidence to support MI for behavioral change, including smoking cessation, in patients with cancer.[363]

In order to promote willingness to quit smoking, the US Preventive Services Task Force (USPSTF) recommends a model of MI that uses the "5 R's" of the personal Relevance of quitting; personal Risks of continuing smoking; personal Rewards of quitting; identifying Roadblocks to quitting; and Repeating the message at every contact.[295] This model encourages that motivational information be relevant to the individual patient, and that clinicians and patients work together to identify personalized risks of smoking and potential rewards of cessation. By having the patient identify perceived roadblocks to quitting, health care professionals can suggest tailored treatments to address patient-reported concerns. Finally, this model recommends repetition of MI at each patient visit, coupled with reminders that repeated quit attempts may be necessary to achieve long-term cessation.

Alternative Treatment Approaches

The Panel has reviewed the available evidence for several alternative smoking cessation treatment approaches. With any approach, it is critical to continue to provide motivational and behavioral support to all patients during quit attempts, regardless of what smoking cessation methods are being used.

There are a wide variety of products on the market that may be used as alternative nicotine products such as non-combustible tobacco products (smokeless tobacco in the form of snus) and nicotine-containing products without tobacco (nicotine pouches). Some of these products have been determined by the FDA's Center for Tobacco Products to be a modified risk tobacco product (MRTP). MRTPs have been determined to be less harmful than cigarette smoking, but MRTPs are not approved for smoking cessation by the FDA's Center for Drug Evaluation and Research (CDER). The long-term health effects of MRTPs remain unknown, but complete transition from cigarette smoking to MRTPs may reduce smoking-related cancer and disease.

The Panel has found insufficient evidence to recommend the use of alternative therapies alone or in combination with standard smoking treatment approaches. Dual use of cigarette smoking and alternative tobacco products may be more harmful than smoking alone and should be actively discouraged. The guidelines recommend that patients use evidence-based cessation methods to avoid any delay in achieving smoking abstinence. Smoking slips and relapses are common, and prior unsuccessful quit attempts with conventional therapies do not justify the use of unproven alternative approaches. For patients who choose to use non-combustible alternative products for smoking cessation, the provider's main focus should be to support complete long-term smoking abstinence.

E-cigarettes are discussed first. See below for a discussion of other alternative approached (eg, exercise, acupuncture, hypnosis).

E-Cigarettes or Vaping

E-cigarettes, often referred to as vaping, are non-combustible tobacco products that deliver nicotine through heated aerosols for inhalation. The popularity of e-cigarettes is a recent phenomenon, and, as such, the available literature is new and relatively limited, particularly within specific

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subpopulations such as patients with cancer. The FDA has authorized the sale of some e-cigarettes as MRTPs, but no e-cigarette has been approved as a smoking cessation device. The FDA also acknowledges the potential for less harm from e-cigarettes compared to combustible cigarettes.[364]

Several health care organizations have released policy statements concerning e-cigarettes, highlighting the urgent need for research on the safety of these devices and efficacy as a cessation aid. In their 2021 statement, the USPSTF concluded that the current evidence is insufficient to recommend e-cigarettes for tobacco cessation and that smoking treatment efforts should continue to focus on behavioral counseling and pharmacotherapy.[365] The American Heart Association encourages use of evidence-based methods for quitting smoking, and states that the long-term health effects of vaping are not well-understood, with emerging evidence suggesting adverse effects on cardiopulmonary health.[366] The American Lung Association also asserts that e-cigarette use is associated with damage to pulmonary health.[367] American Cancer Society (ACS), AACR, and ASCO recognize the potential for e-cigarettes to alter existing smoking behaviors, as well as the lack of definitive data regarding associated benefits and harms.[368,369] Concurrent use of both e-cigarettes and combustible cigarettes is strongly discouraged by the ACS.[370]

To summarize the current positions of leading medical and oncology organizations, experts in the field generally acknowledge that e-cigarettes may offer an attractive approach for smoking cessation in certain populations. However, the policy statements described above also highlight the unknown potential for e-cigarettes to affect symptoms associated with nicotine use disorder, combustible tobacco product use, and renormalization of smoking behaviors.

Efficacy and Effectiveness

Randomized Controlled Trials

In a 2024 RCT, individuals who wanted to quit combustible cigarettes (N = 1246) were randomized to receive e-cigarettes in addition to standard smoking treatment, or standard treatment without e-cigarettes.[371] Continuous abstinence from smoking combustible cigarettes at 6 months was significantly greater in the e-cigarette intervention arm, compared to the control arm (28.9% vs. 16.3%, respectively; relative risk, 1.77; 95% CI, 1.43–2.20). However, it is important to note that, among the intervention arm (n = 552), about half (48.4%) continued to use e-cigarettes at the time of 6-month follow-up.

In the first head-to-head comparison, an RCT (n = 886) compared e-cigarettes to NRT (combination NRT allowed), paired with at least 4 weeks of behavioral support.[372] Abstinence rates at 1 year were 18% and 9.9% for e-cigarettes and NRT, respectively (RR, 1.83; 95% CI, 1.30–2.58; $P <$ .001). However, prolonged e-cigarette use extended beyond the initial treatment period, particularly among abstainers, with 80% continuing to use e-cigarettes after 1 year, compared to 9% who continued to use NRT.[373] Further, one in four participants in the e-cigarette arm did not quit combustible cigarettes, meaning the strategy of e-cigarette for smoking cessation may lead to dual use.[374]

In a more recent single-center RCT from northern Finland, participants who smoked daily (N = 458) were randomized to receive12 weeks of nicotine-containing e-cigarettes with placebo tablets, varenicline with nicotine-free e-cigarettes, or placebo tablets with nicotine-free e-cigarettes.[375] All participants received motivational interviewing. Though 7-day smoking abstinence rates at week 26 (confirmed by exhaled carbon monoxide) were significantly greater ($P <$.001) in both the nicotine-containing e-cigarettes (40.4%) and varenicline (43.8%) arms, compared

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to the placebo arm (19.7%), differences between the e-cigarette and varenicline arms were not statistically significant.

In a secondary analysis from a randomized trial in which intensive smoking treatment was compared to standard treatment in patients with cancer who smoke cigarettes (N = 303), e-cigarette use was not significantly associated with 6-month abstinence from cigarettes, even when controlling for tobacco treatment study group (OR, 0.42; 95% CI, 0.13–1.31).[376]

Cochrane Reviews

The efficacy of e-cigarettes for quitting combustible cigarettes have been evaluated in Cochrane reviews. A 2023 review showed that nicotine e-cigarettes were associated with higher quit rates, compared to control, with high-certainty evidence (OR, 2.37; 95% CI, 1.73–3.24; 16 RCTs with 3828 participants).[377] This OR is comparable to that for varenicline and combination NRT. This review showed that quit rates for EC were greater than single forms of NRT, bupropion, and nortriptyline. A 2024 review including 88 studies (47 RCTs) showed that nicotine e-cigarettes were associated with greater quit rates compared to e-cigarettes without nicotine (high-certainty evidence).[292] This review also showed that nicotine e-cigarettes were associated with quit rates comparable to that for NRT (moderate-certainty evidence). Another 2024 review showed that e-cigarettes were associated with greater quit rates (as assessed by biochemical validation) than NRT (RR, 1.59; 95% CI, 1.29–1.93) with high certainty of evidence.[378] However, this analysis did not distinguish between combination NRT and single forms of NRT. A 2022 review also found better abstinence from smoking rates with nicotine e-cigarettes, compared to NRT, with high certainty (RR, 1.63; 95% CI, 1.30–2.04).[379]

Safety

Despite the compelling data on efficacy, the decision to recommend e-cigarettes as a tool to help quit combustible cigarettes cannot be based on efficacy data alone; the safety of e-cigarette use must also be considered.

Secondary analyses from a 2022 Cochrane review (pooled data from three studies) showed a decrease in carbon monoxide with e-cigarette use, compared to NRT.[379] A 2024 random-effects meta-analysis was the first published direct comparison of the health effects of e-cigarette or dual use with combustible cigarettes in the general public.[380] When comparing combustible cigarettes to e-cigarettes, lower disease risk was observed for e-cigarettes for asthma (OR, 0.84; 95% CI, 0.74–0.95), chronic obstructive pulmonary disease (OR, 0.53; 95% CI, 0.38–0.74), and oral disease (OR, 0.87; 95% CI, 0.76–1.00). There was no significant difference in risk observed for cardiovascular disease, stroke, and metabolic dysfunction. When comparing dual use to cigarette use, there was a greater risk observed for all health outcomes, with ORs ranging from 1.20 to 1.41. When comparing e-cigarette use and dual use to non-use of either product, greater risk for all health outcomes was also observed (ORs ranging from 1.24 to 1.47 for e-cigarette use, and 1.49 to 3.29 for dual use). There were some limitations of this study. First, there was no analysis of the specific type of device or liquids used for e-cigarettes. Second, frequency or intensity of use of the two products was not examined, so dose-response relationships could not be assessed. Finally, most of the included studies were cross-sectional. Nevertheless, these findings are consistent with what basic science has demonstrated; that e-cigarette aerosols may cause DNA damage and may be procarcinogenic.[381]

To summarize, though e-cigarettes are associated with less harm compared to combustible cigarettes, there is evidence that “incomplete switching” (ie, dual use) is associated with greater harm than use of

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combustible cigarettes alone.[381] Also, there are currently no data on the interaction of e-cigarette use or dual use on cancer treatment or prognosis.[369] Even though e-cigarettes alone are associated with less harm than combustible cigarettes, they are not harmless, with dual use being particularly harmful. The long-term health effects of vaping are unknown. To date, there are no guidelines for e-cigarette cessation, though one protocol has been developed for use of varenicline for this purpose.[382]

An outbreak of significant respiratory illness was observed in patients who vaped e-cigarettes and illicit drugs, so called e-cigarette or vaping use-associated lung injury (EVALI). EVALI is associated with respiratory, constitutional, and gastrointestinal symptoms,[383] with about half of patients experiencing EVALI requiring intensive care for respiratory failure.[384] Factors correlated with EVALI-associated death include age ≥35 years, obesity, and the presence of a chronic medical condition (ie, chronic respiratory disease and cardiac disease).[385] EVALI was ultimately determined to be associated with additives in e-liquids for illicit drug use, and not due to commercial e-cigarettes. An analysis of bronchoalveolar lavage (BAL) fluid in 51 patients with EVALI and 99 patients without EVALI (including patients who never smoked, patients who smoked combustible cigarettes, and patients who vaped e-cigarettes) showed that vitamin E acetate was found in 94% of patients with EVALI and in none of the comparators, making it the likely causative agent.[386] Limonene and coconut oil were found in the BAL fluid of two patients with EVALI (1 patient each), but no other potentially toxic substances (eg, plant oil, petroleum distillates) were found. The reporting structure for these events has been officially closed. Vitamin E acetate has largely been eliminated from vaping products in the United States.

Other Studies

Cross-sectional studies have evaluated patient's e-cigarette use, but these studies provide insufficient evidence about efficacy and safety.

One study examined e-cigarette use in 1074 patients with cancer who enrolled in a tobacco treatment program at a comprehensive cancer center.[387] The study revealed a marked increase in e-cigarette use from 10.6% to 38.5% between 2012 and 2013. Patients who used e-cigarettes were most often diagnosed with thoracic or head and neck cancers, were more nicotine dependent, and had greater numbers of prior quit attempts. At follow-up (6–12 months after intake), patients who use e-cigarettes were no more likely to have quit than patients who did not use e-cigarettes (OR, 1.0; 95% CI, 0.5–1.7). Two cross-sectional studies of cancer survivors in the United States showed that e-cigarette use was highest among patients who currently smoke cigarettes, compared with patients who formerly smoked or who never smoked cigarettes.[388,389] Another cross-sectional study surveyed 121 patients with cancer who were currently using e-cigarettes.[390] Eighty-one percent of these surveyed patients who use e-cigarettes reported smoking cessation as their motivation for initiating e-cigarette use, but 51% also reported dual use of combustible cigarettes with e-cigarettes. Additionally, more than 70% reported that they had not discussed their e-cigarette use with their oncology care providers.

A prospective cohort study compared e-cigarette use, tobacco use, and dual use with longer-term follow-up. At 24 months, those who used only e-cigarettes were more likely to remain abstinent from other tobacco products than those who used only tobacco or those who used both e-cigarettes and tobacco (abstinence rates of 61.1%, 23.1%, and 26.0%, respectively).[391] Analyses of adults who used e-cigarettes and enrolled in the Population Assessment of Tobacco and Health Study (n = 2835), an ongoing longitudinal cohort study, showed that 48.8% had quit both e-

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cigarettes and combustible cigarettes over a 1-year period (2013–2014 and 2014–2015), while 11.4% decreased their frequency of e-cigarette use.[392] However, it is important to note that there are now more potent e-cigarette products available. Therefore, these results may not be generalizable to the present day. The same study showed that, among baseline dual users (n = 2036), 87.8% reported continued smoking of combustible cigarettes at the time of follow-up assessment. Among those who smoked solely e-cigarettes at baseline (n = 869), 43.4% continued to smoke e-cigarettes after 1 year, while 30.9% had quit smoking e-cigarettes. Non-daily e-cigarette use at baseline was associated with discontinued e-cigarette use at the time of follow-up assessment. However, those who used e-cigarettes daily at baseline were more likely to report having quit combustible cigarettes after 1 year, compared to those with non-daily use. Similar associations were found in a large cohort study of individuals who smoke initially not using e-cigarettes and with no plans to ever quit smoking.[393] Abstinence rates of cigarette use after 12 months were higher in those who used e-cigarettes daily, compared to those who did not (28.0% vs. 5.8%, respectively; OR, 8.11; 95% CI, 3.14–20.97). However, non-daily e-cigarette use was not associated with abstinence from cigarettes (OR, 0.53; 95% CI, 0.08–3.35).

A large cross-sectional survey of 5863 adults in the United Kingdom assessed the “real-world effectiveness” of e-cigarettes for smoking cessation compared to NRT and unaided quitting, revealing that those who use e-cigarettes were more likely to self-report abstinence from cigarettes compared with the other cohorts (e-cigarettes vs. NRT: OR, 2.23; 95% CI, 1.70–2.93; e-cigarettes vs. no aid: OR, 1.38; 95% CI, 1.08–1.76).[394] However, it is important to note that e-cigarettes in the United Kingdom are different than those in the United States, which lags in product regulation in the real-world marketplace.

E-cigarette use did not decrease cigarette smoking in a cross-sectional study in 106 patients with head and neck cancers seeking to quit. Those who abstained from using e-cigarettes had a significantly greater rate of cessation compared with patients who use e-cigarettes (72% vs. 39%; P = .0057).[395]

NCCN Recommendations

At the current time, e-cigarettes are not an FDA-approved method for smoking treatment. It is not yet clear how factors such as real-world use or the addition of behavior therapy may influence the efficacy, safety, and effectiveness of e-cigarettes. Patients should be counseled toward the use of evidence-based smoking treatment approaches. For patients who choose to use e-cigarettes for smoking cessation, despite recommendations to use evidence-based methods, encourage combustible cigarette smoking abstinence even if this is by the use of e-cigarettes, and incorporate behavioral counseling, given the harm to one’s health that is associated with smoking combustible cigarettes.

Initiation of e-cigarette use may lead to long-term use and the continued dual use of both e-cigarettes and cigarettes, which has greater health risk concerns. Dual use of smoking cigarettes and e-cigarettes is associated with high exposure to tobacco toxicants.[396,397] For patients who use both e-cigarettes and combustible tobacco, dual use is strongly discouraged, as it continues to pose a risk of cancer and other smoking-related diseases.[398-400] When the patient develops more confidence about not smoking, cessation of e-cigarettes should be encouraged, but not at the risk of relapse to smoking combustible products.

Optimal methods for vaping cessation are currently unknown. Because of this, providers are encouraged to following guidelines for smoking cessation in the treatment of e-cigarette use, including behavioral counseling and pharmacotherapy.

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Other Alternative Approaches

Very limited data exist to support exercise-based interventions; small study size, inadequate controls, and insufficient exercise intensity limit the ability to make conclusions based on the existing evidence.[401] Sufficient efficacy data are also lacking to support the use of alternative therapies such as acupuncture, hypnosis, and nutritional supplements. A 2014 systematic review of the data on acupuncture, acupressure, and laser therapy revealed no consistent, bias-free evidence to support these methods for smoking cessation, although pooled evidence was suggestive of possible short-term benefits.[402] Acupuncture was less effective than NRT and there was no evidence to support electrostimulation for smoking cessation. Similarly, systematic reviews of the data on hypnosis for smoking cessation revealed inadequate high-quality evidence to support this approach.[403,404] Claims of efficacy data for hypnosis from several studies were not substantiated by the review of RCT data. Controlled studies are needed to provide higher quality evidence on these interventions both in the general population and among patients with cancer.

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Discussion update in progress

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